Statistical Learning

Consolidated and Expanded Lecture Notes

Compiled from Lectures 4–14 and Tutorials 1, 3

Ridge & LASSO regression · Elastic Net · Splines & projection pursuit · Neural networks · Kernel and k-NN regression · Regression & classification trees · Cost-complexity pruning · Bagging & random forests · MARS · Statistical decision theory & the Bayes classifier · Robust regression (LAD, IRWLS)

Contents

Part I: Regularized Linear Models and Basis Expansions

1 Multicollinearity and the Case for Regularization

1.1 The problem

Consider the standard linear model $\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{X} \in \mathbb{R}^{n\times(p+1)}$ (first column all ones), with OLS estimator

$$\widehat{\beta}_{LS} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}, \qquad \mathrm{Var}(\widehat{\beta}_{LS}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}.$$

When predictors are highly correlated, $\boldsymbol{X}'\boldsymbol{X}$ is nearly singular: its smallest eigenvalues $\sigma_j^2 \approx 0$. Since $\mathrm{Var}(\widehat{\beta}_{LS}) = \sigma^2 \sum_j v_j v_j'/\sigma_j^2$ (writing the eigendecomposition $\boldsymbol{X}'\boldsymbol{X} = \sum_j \sigma_j^2 v_j v_j'$), directions with small σ_j^2 inflate the variance of $\widehat{\beta}_{LS}$ enormously. Coefficients become large, unstable, and sign-flip under small data perturbations, even though predictions $\boldsymbol{X}\widehat{\beta}_{LS}$ may still fit reasonably well. Two broad remedies exist: *subset selection* (discrete, remove variables) and *shrinkage / regularization* (continuous, shrink coefficients).

1.2 Subset selection

Forward selection or backward deletion produce a *nested* sequence of models (each differing from its neighbour by one variable). A single member of this sequence is then chosen by a criterion that trades off fit against complexity:

- **Adjusted** R^2: $1 - \dfrac{RSS/(n-p-1)}{TSS/(n-1)}$, penalizes adding variables that do not reduce RSS enough to compensate for the lost degree of freedom.
- **Mallow's** C_p: $C_p = \dfrac{RSS_p}{\hat{\sigma}^2} - n + 2p$, an (asymptotically) unbiased estimate of expected out-of-sample prediction error, up to constants.
- **AIC** $= -2\log L + 2k$, **BIC** $= -2\log L + k\log n$ — BIC penalizes complexity more heavily for large n and is model-selection consistent under regularity conditions, whereas AIC is efficient (asymptotically minimizes prediction risk) but not consistent.

Subset selection is a discrete, combinatorial procedure (it either keeps or drops a variable) and is therefore high-variance itself: small perturbations of the data can change which subset is chosen. This motivates continuous shrinkage.

2 Ridge Regression

2.1 Formulation

Ridge regression shrinks coefficients continuously rather than deleting them. It solves the constrained problem

$$\min_{\boldsymbol{\beta}} \|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|^2 \le c,$$

which, by Lagrangian duality, is equivalent (for a corresponding $\lambda > 0$) to the unconstrained penalized problem

$$\Psi(\boldsymbol{\beta}) = \|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2 \longrightarrow \min_{\boldsymbol{\beta}}.$$

Proof

Derivation of the ridge estimator. Expanding,

$$\Psi(\boldsymbol{\beta}) = \boldsymbol{Y}'\boldsymbol{Y} - 2\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{Y} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} + \lambda\boldsymbol{\beta}'\boldsymbol{\beta}.$$

Differentiating and setting the gradient to zero,

$$\nabla\Psi(\boldsymbol{\beta}) = -2\boldsymbol{X}'\boldsymbol{Y} + 2\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} + 2\lambda\boldsymbol{\beta} = 0 \implies (\boldsymbol{X}'\boldsymbol{X} + \lambda I)\boldsymbol{\beta} = \boldsymbol{X}'\boldsymbol{Y},$$

so that

$$\boxed{\widehat{\beta}_{\text{Ridge}} = (\boldsymbol{X}'\boldsymbol{X} + \lambda I)^{-1}\boldsymbol{X}'\boldsymbol{Y}.}$$

The Hessian of Ψ is $2(\boldsymbol{X}'\boldsymbol{X} + \lambda I) \succeq 2\lambda I \succ 0$ for $\lambda > 0$, so Ψ is strictly convex and this stationary point is the unique global minimizer. Crucially, $\boldsymbol{X}'\boldsymbol{X} + \lambda I$ is *always* invertible for $\lambda > 0$ (its eigenvalues are $\sigma_j^2 + \lambda \geq \lambda > 0$), even when $\boldsymbol{X}'\boldsymbol{X}$ itself is singular. This is why ridge stabilizes estimation under multicollinearity or when $p > n$.

2.2 Geometric interpretation

In the two-predictor case, the constraint $\beta_1^2 + \beta_2^2 \leq c$ is a disk, while the contours of the RSS $\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|^2$ around $\widehat{\beta}_{LS}$ are concentric ellipses (governed by $\boldsymbol{X}'\boldsymbol{X}$). The ridge solution is the point where the smallest ellipse touches the disk — geometrically pulled from $\widehat{\beta}_{LS}$ toward the origin along a direction that is generally *not* axis-aligned, so ridge shrinks coefficients but essentially never sets any exactly to zero.

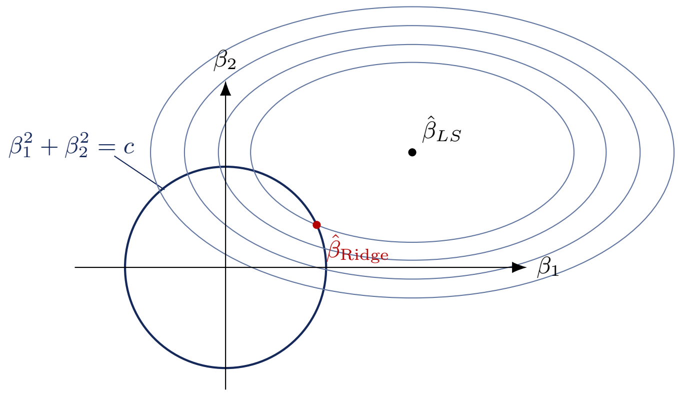

Figure 1: Ridge estimate as the tangency point between elliptical RSS contours (centered at $\hat{\beta}_{LS}$) and the circular ℓ_2 constraint region. The innermost ellipse shown is exactly tangent to the constraint circle at $\hat{\beta}_{\text{Ridge}}$.

2.3 A key algebraic identity: decomposing the RSS surface

Proposition 2.1. *For any* $\boldsymbol{\beta}$,

$$\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 = \|\boldsymbol{Y} - \boldsymbol{X}\widehat{\beta}_{LS}\|^2 + (\boldsymbol{\beta} - \widehat{\beta}_{LS})'\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{\beta} - \widehat{\beta}_{LS}).$$

Proof

Write $\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta} = (\boldsymbol{Y} - \boldsymbol{X}\widehat{\beta}_{LS}) + \boldsymbol{X}(\widehat{\beta}_{LS} - \boldsymbol{\beta})$. Expanding the squared norm,

$$\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 = \|\boldsymbol{Y} - \boldsymbol{X}\widehat{\beta}_{LS}\|^2 + 2(\boldsymbol{Y} - \boldsymbol{X}\widehat{\beta}_{LS})'\boldsymbol{X}(\widehat{\beta}_{LS} - \boldsymbol{\beta}) + \|\boldsymbol{X}(\widehat{\beta}_{LS} - \boldsymbol{\beta})\|^2.$$

The cross term vanishes because $\boldsymbol{X}'(\boldsymbol{Y} - \boldsymbol{X}\widehat{\beta}_{LS}) = \boldsymbol{X}'\boldsymbol{Y} - \boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} = 0$ (normal equations), and $\|\boldsymbol{X}(\widehat{\beta}_{LS} - \boldsymbol{\beta})\|^2 = (\boldsymbol{\beta} - \widehat{\beta}_{LS})'\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{\beta} - \widehat{\beta}_{LS})$.

This shows the RSS surface is exactly the minimum RSS plus a quadratic penalty term whose level sets $(\boldsymbol{\beta} - \widehat{\beta}_{LS})'\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{\beta} - \widehat{\beta}_{LS}) = \text{const}$ are precisely the ellipses of Figure 1 (governed by $\boldsymbol{X}'\boldsymbol{X}$, hence generally not circular).

2.4 Bias, variance, and the bias–variance trade-off

Since $\widehat{\beta}_{\text{Ridge}} = (\boldsymbol{X}'\boldsymbol{X} + \lambda I)^{-1}\boldsymbol{X}'\boldsymbol{Y}$ and $\mathbb{E}[\boldsymbol{Y}] = \boldsymbol{X}\boldsymbol{\beta}$,

$$\mathbb{E}[\widehat{\beta}_{\text{Ridge}}] = (\boldsymbol{X}'\boldsymbol{X} + \lambda I)^{-1}\boldsymbol{X}'\boldsymbol{X}\,\boldsymbol{\beta} \neq \boldsymbol{\beta} \quad (\lambda > 0),$$

so ridge is *biased*. Its covariance is

$$\mathrm{Var}(\widehat{\beta}_{\text{Ridge}}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X}+\lambda I)^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X}+\lambda I)^{-1} \preceq \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} = \mathrm{Var}(\widehat{\beta}_{LS}),$$

i.e. ridge trades bias for a reduction in variance. For a suitably chosen λ, the total mean-squared error $\mathbb{E}\|\widehat{\beta}_{\text{Ridge}} - \boldsymbol{\beta}\|^2$ can be strictly smaller than that of OLS, especially when predictors are collinear (small σ_j^2 directions dominate OLS variance).

2.5 Singular value decomposition and coordinate-wise shrinkage

Let the centered design $\boldsymbol{X}_c = UDV'$ (SVD), with $U'U = I_p$, $V'V = VV' = I_p$, $D = \mathrm{diag}(d_1, \ldots, d_p)$. Then

$$\widehat{\beta}_{\text{Ridge}}(\lambda) = V(D^2+\lambda I)^{-1}DU'\boldsymbol{y}_c = \sum_{j=1}^{p} v_j \left(\frac{d_j}{d_j^2+\lambda}\right) u_j'\boldsymbol{y}_c, \qquad \widehat{\beta}_{LS} = \sum_{j=1}^{p} v_j \left(\frac{1}{d_j}\right) u_j'\boldsymbol{y}_c.$$

Ridge therefore shrinks the j-th principal-component projection of the OLS fit by the factor $d_j^2/(d_j^2+\lambda) \in (0,1)$: directions with small singular value d_j (i.e. low-variance / collinear directions) are shrunk the most, exactly the directions responsible for OLS's variance blow-up. This gives the **effective degrees of freedom**

$$\mathrm{df}(\lambda) = \mathrm{tr}(\boldsymbol{X}_c(\boldsymbol{X}_c'\boldsymbol{X}_c+\lambda I)^{-1}\boldsymbol{X}_c') = \sum_{j=1}^{p}\frac{d_j^2}{d_j^2+\lambda},$$

which decreases smoothly from p (at $\lambda = 0$) to 0 (as $\lambda \to \infty$).

2.6 Choosing λ

λ and the constraint budget c are in one-to-one, monotonically decreasing correspondence. In practice λ is chosen by *cross-validation*: fit ridge on training folds over a grid of λ values, evaluate held-out prediction error, and pick the λ minimizing cross-validated error (often the "1-SE rule" is used, choosing the largest λ within one standard error of the minimum, for parsimony).

2.7 Coordinate dependence and standardization

Unlike OLS, ridge is *not* invariant to the coordinate system in which predictors are recorded.

Proposition 2.2 (OLS invariance under $\boldsymbol{X} \mapsto \boldsymbol{X}A$)**.** *For any invertible $A \in \mathbb{R}^{p\times p}$, the OLS projection (hat) matrix satisfies $H_{XA} = H_X$, so fitted values, residuals, R^2 are unchanged, and $\widehat{\beta}_{XA} = A^{-1}\widehat{\beta}_X$.*

Proof

$H_{XA} = XA\big((XA)'XA\big)^{-1}(XA)' = XA\,A^{-1}(X'X)^{-1}(A')^{-1}A'X' = X(X'X)^{-1}X' = H_X$. Also $\widehat{\beta}_{XA} = ((XA)'XA)^{-1}(XA)'y = A^{-1}(X'X)^{-1}X'y = A^{-1}\widehat{\beta}_X$.

For ridge, transforming $\boldsymbol{X}_c \mapsto \boldsymbol{X}_cA$ gives $\widehat{\beta}^{\text{Ridge}}_{XA}(\lambda) = (A'X_c'X_cA+\lambda I)^{-1}A'X_c'y_c$; because $A'X_c'X_cA+\lambda I = A'(X_c'X_c + \lambda(AA')^{-1})A$, the identity penalty λI prevents A from factoring out as it did for OLS — the penalty would need to be $\lambda(AA')^{-1}$ to recover the transformed solution. Consequently:

- Ridge on *raw* $\boldsymbol{X}$ is not scale invariant: a predictor recorded in smaller units needs a larger β_j for the same effect and is thus penalized more, distorting relative shrinkage.
- **Standardizing** each predictor, $z_{ij} = (x_{ij} - \bar{x}_j)/s_j$, restores invariance to arbitrary re-scaling (diagonal $A = D$), because the standardized coefficients are unit-free.
- Standardization does *not* restore invariance to general rotations/shears (A non-diagonal): the ℓ_2 ball is rotationally symmetric only in the standardized coordinate axes chosen, and correlated predictors still interact with the isotropic penalty in a basis-dependent way — full invariance would require the (data-dependent) whitening transform $A = VD^{-1}$ from the SVD above.

The standard convention is therefore: **never penalize the intercept** (center y and each column of X first, fit the penalized model on centered data without an intercept, then recover $\hat{\alpha} = \bar{y} - \bar{x}'\widehat{\beta}$), and standardize predictors before penalizing.

3 The LASSO

3.1 Objective and the orthonormal case

The Least Absolute Shrinkage and Selection Operator (LASSO) replaces the ℓ_2 penalty with an ℓ_1 penalty:

$$\Delta(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big(y_i - \sum_{j=1}^{p} \beta_j x_{ji}\Big)^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

To build intuition, first suppose the design is orthonormal, $\boldsymbol{X}'\boldsymbol{X} = I$, so that $\widehat{\beta}_{LS} = \boldsymbol{X}'\boldsymbol{Y}$, with j-th coordinate $\widehat{\beta}_{j,LS} = \sum_i y_i x_{ji}$. Because $|\beta_j|$ is non-differentiable at 0, we treat the cases $\beta_k > 0$, $\beta_k < 0$, $\beta_k = 0$ separately.

Case $\beta_k > 0$. Δ is differentiable in β_k:

$$\frac{\partial \Delta}{\partial \beta_k} = -2 \sum_i \Big(y_i - \sum_j \beta_j x_{ji}\Big) x_{ki} + \lambda = 0.$$

Using $\boldsymbol{X}'\boldsymbol{X} = I$ (cross terms vanish), this reduces to $-\widehat{\beta}_{k,LS} + \beta_k + \lambda/2 = 0$, i.e. $\beta_k = \widehat{\beta}_{k,LS} - \lambda/2$, valid only when this is indeed > 0, i.e. when $\widehat{\beta}_{k,LS} > \lambda/2$.

Case $\beta_k < 0$. By the symmetric argument, $\beta_k = \widehat{\beta}_{k,LS} + \lambda/2$, valid when $\widehat{\beta}_{k,LS} < -\lambda/2$.

Case $\beta_k = 0$ (subgradient). The subdifferential of Δ in β_k at 0 is

$$\partial \Delta(0) = -2\widehat{\beta}_{k,LS} + [-\lambda, \lambda],$$

and $\beta_k = 0$ is optimal iff $0 \in \partial\Delta(0)$, i.e. $|\widehat{\beta}_{k,LS}| \le \lambda/2$.

Soft-thresholding solution (orthonormal design)

Writing $\lambda^* = \lambda/2$,

$$\widehat{\beta}_{k,\text{LASSO}} = \begin{cases} \widehat{\beta}_{k,LS} - \lambda^* & \widehat{\beta}_{k,LS} > \lambda^* \\ 0 & |\widehat{\beta}_{k,LS}| \le \lambda^* \\ \widehat{\beta}_{k,LS} + \lambda^* & \widehat{\beta}_{k,LS} < -\lambda^* \end{cases} = \operatorname{sign}(\widehat{\beta}_{k,LS})\big(|\widehat{\beta}_{k,LS}| - \lambda^*\big)_+ \; =: \; S_{\lambda^*}(\widehat{\beta}_{k,LS}).$$

Ridge vs LASSO mechanics

Ridge *multiplies*: $\widehat{\beta}_k^{\text{Ridge}} = \widehat{\beta}_{k,LS}/(1+\lambda)$ — a positive scaling never lands exactly on zero for finite λ. LASSO *subtracts*: $\operatorname{sign}(\widehat{\beta}_{k,LS})(|\widehat{\beta}_{k,LS}| - \lambda^*)_+$ — a whole interval $[-\lambda^*, \lambda^*]$ collapses to zero. This is the origin of LASSO's simultaneous shrinkage *and* variable selection, which ridge lacks.

3.2 The general case ($X'X \neq I$): KKT / subgradient conditions

For a general design, the coordinates couple through the off-diagonal entries of $\boldsymbol{X}'\boldsymbol{X}$ and no closed form exists. Writing the (sample-averaged) objective $L(\boldsymbol{\beta}) = \frac{1}{2n}\|\boldsymbol{y}_c - \boldsymbol{X}_c\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1$, stationarity requires $0 \in \nabla\big(\frac{1}{2n}\|\boldsymbol{y}_c - \boldsymbol{X}_c\boldsymbol{\beta}\|^2\big) + \lambda\,\partial\|\boldsymbol{\beta}\|_1$, i.e. component-wise, with x_j the j-th column of $\boldsymbol{X}_c$,

$$\frac{1}{n} x_j'(\boldsymbol{y}_c - \boldsymbol{X}_c\widehat{\beta}) = \lambda \operatorname{sign}(\widehat{\beta}_j) \text{ if } \widehat{\beta}_j \neq 0, \qquad \left|\frac{1}{n} x_j'(\boldsymbol{y}_c - \boldsymbol{X}_c\widehat{\beta})\right| \le \lambda \text{ if } \widehat{\beta}_j = 0.$$

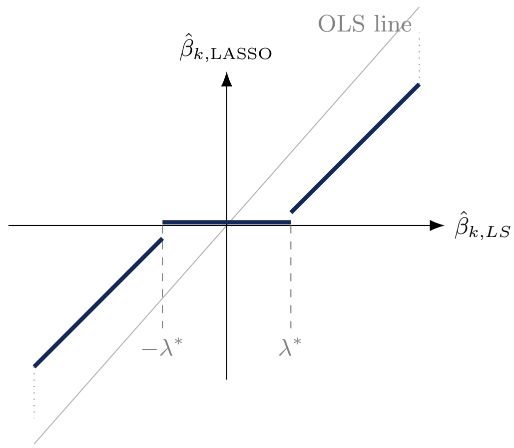

Figure 2: LASSO soft-thresholding: flat (exactly zero, bold) on $[-\lambda^*, \lambda^*]$, then a unit-slope translation of the OLS estimate (thin diagonal reference line) by λ^* toward zero.

That is: a variable stays out of the model ($\widehat{\beta}_j = 0$) exactly when its empirical correlation with the current residual does not exceed λ in absolute value. Because the problem remains convex (sum of a convex quadratic and a convex, non-smooth penalty), the solution set is characterized exactly by these KKT conditions, and can be computed via **coordinate descent** (cycling through a generalization of the soft-threshold update above holding other coordinates fixed), **quadratic programming**, or path algorithms such as **LARS** (Least Angle Regression), which traces the entire piecewise-linear solution path in λ.

3.3 Constrained vs. penalized formulations

The Lagrangian form $\min_\beta RSS(\beta) + \lambda\|\beta\|_1$ and the constrained (budget) form $\min_\beta RSS(\beta)$ s.t. $\|\beta\|_1 \le t$ are equivalent in the sense that for every $\lambda \ge 0$ there is a $t(\lambda)$ solving the same problem, and vice versa.

Proof

($\Rightarrow$) Let $\widehat{\beta}$ minimize $RSS(\beta) + \lambda\|\beta\|_1$ and set $t = \|\widehat{\beta}\|_1$. For any β with $\|\beta\|_1 \le t$, optimality of $\widehat{\beta}$ gives $RSS(\beta) + \lambda\|\beta\|_1 \ge RSS(\widehat{\beta}) + \lambda\|\widehat{\beta}\|_1$, so $RSS(\beta) \ge RSS(\widehat{\beta}) + \lambda(\|\widehat{\beta}\|_1 - \|\beta\|_1) \ge RSS(\widehat{\beta})$ since $\|\beta\|_1 \le t = \|\widehat{\beta}\|_1$. Hence $\widehat{\beta}$ solves the constrained problem with budget t. ($\Leftarrow$) RSS is convex and $g(\beta) = \|\beta\|_1 - t$ is convex; for $t > 0$, $\beta = 0$ satisfies $g(0) < 0$ (Slater's condition), so strong duality holds and KKT gives a multiplier $\lambda \ge 0$ with $0 \in \partial_\beta\big(RSS(\widehat{\beta}) + \lambda(\|\widehat{\beta}\|_1 - t)\big)$ and complementary slackness $\lambda(\|\widehat{\beta}\|_1 - t) = 0$, i.e. $\widehat{\beta}$ solves the Lagrangian problem.

3.4 Path geometry

As λ decreases from ∞ to 0, LASSO coefficients trace a continuous, *piecewise-linear* path in λ, with coordinates entering/leaving the active set at discrete kinks — this is exploited by LARS to compute the whole path in $O(p^2 n)$-type cost. Since $\|A\beta\|_1 = \|\beta\|_1$ iff A is a signed permutation matrix, LASSO solutions are invariant to reordering/sign-flipping predictors but *not* to general rotations (a rotation moves the ℓ_1 ball's corners off the coordinate axes, changing which variables are selected).

4 The Elastic Net

4.1 Motivation

LASSO has two practical weaknesses: (i) among a group of highly correlated predictors it tends to pick one arbitrarily and zero out the rest (unstable under resampling); (ii) when $p > n$, the convex geometry of the ℓ_1 ball caps the number of selected variables at n. Zou and Hastie (2005) proposed combining ℓ_1 and ℓ_2 penalties:

$$\widehat{\beta}_{\text{enet}}(\lambda, \alpha) = \arg\min_{\boldsymbol{\beta}} \left\{ \tfrac{1}{2n}\|\boldsymbol{y}_c - \boldsymbol{X}_c\boldsymbol{\beta}\|_2^2 + \lambda\Big(\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2\Big) \right\}, \qquad \alpha \in [0, 1],$$

with $\alpha = 1$ recovering LASSO and $\alpha = 0$ recovering ridge.

4.2 The grouping effect

Theorem 4.1 (Zou–Hastie grouping bound). *Let predictors be standardized so $x_i'x_i = x_j'x_j = 1$ with correlation $\rho = x_i'x_j$, and let $\lambda_2 = \lambda(1-\alpha)$. If $\widehat{\beta}_i\widehat{\beta}_j > 0$,*

$$\frac{|\widehat{\beta}_i - \widehat{\beta}_j|}{\|\boldsymbol{y}_c\|_1} \le \frac{1}{\lambda_2}\sqrt{2(1-\rho)}.$$

Proof

The subgradient stationarity conditions for coordinates i, j are

$$-\tfrac{1}{n}x_i'(\boldsymbol{y}_c - \boldsymbol{X}_c\widehat{\beta}) + \lambda\alpha\,\mathrm{sign}(\widehat{\beta}_i) + \lambda_2\widehat{\beta}_i = 0, \qquad -\tfrac{1}{n}x_j'(\boldsymbol{y}_c - \boldsymbol{X}_c\widehat{\beta}) + \lambda\alpha\,\mathrm{sign}(\widehat{\beta}_j) + \lambda_2\widehat{\beta}_j = 0.$$

Since $\widehat{\beta}_i\widehat{\beta}_j > 0$, $\mathrm{sign}(\widehat{\beta}_i) = \mathrm{sign}(\widehat{\beta}_j)$, so subtracting cancels the ℓ_1 terms:

$$\lambda_2(\widehat{\beta}_i - \widehat{\beta}_j) = \tfrac{1}{n}(x_i - x_j)'\hat{r}, \qquad \hat{r} = \boldsymbol{y}_c - \boldsymbol{X}_c\widehat{\beta}.$$

By Cauchy–Schwarz, $\lambda_2|\widehat{\beta}_i - \widehat{\beta}_j| \le \frac{1}{n}\|x_i - x_j\|_2\|\hat{r}\|_2$. Now $\|x_i - x_j\|_2^2 = x_i'x_i - 2x_i'x_j + x_j'x_j = 2(1-\rho)$, and because $\widehat{\beta}$ minimizes the objective (which is $\le$ its value at $\beta = 0$), $\|\hat{r}\|_2 \le \|\boldsymbol{y}_c\|_2 \le \|\boldsymbol{y}_c\|_1$. Combining gives the stated bound.

As $\rho \to 1$ the bound $\to 0$: highly correlated predictors are forced to receive nearly identical coefficients — the "grouping" effect that pure LASSO lacks.

4.3 Why the elastic net still performs selection

The stationarity condition for $\widehat{\beta}_j = 0$ is $\left|\frac{1}{n}x_j'(\boldsymbol{y}_c - \boldsymbol{X}_c\widehat{\beta})\right| \le \lambda\alpha$; the smooth ℓ_2 term contributes zero gradient at $\beta_j = 0$, so the width of the zero-region is generated entirely by the ℓ_1 part and is strictly positive whenever $\alpha > 0$. Geometrically, the constraint region $\{\beta : \alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2 \le s\}$ has non-differentiable corners on the coordinate axes for every $\alpha > 0$ (the interior vertex angle there is $\theta = \arccos\left(\frac{\alpha^2-1}{\alpha^2+1}\right)$, equal to 90° at $\alpha = 1$ and widening to 180°, i.e. smooth, only as $\alpha \to 0$), so exact zeros persist for any $\alpha > 0$.

Property	OLS	Ridge	LASSO	Elastic Net
Penalty	none	$\lambda\|\beta\|_2^2$	$\lambda\|\beta\|_1$	$\lambda[\alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2]$
Closed form	$(X'X)^{-1}X'y$	$(X'X + \lambda I)^{-1}X'y$	no (except orthonormal)	no (except orthonormal)
Defined for $p > n$	no	yes	yes	yes
Uniqueness	unique if $p \le n$	always unique	$\hat{y}$ unique, $\widehat{\beta}$ may not be	always unique ($\alpha < 1$)
Shrinkage	none	multiplicative $b_j/(1+\lambda)$	soft-threshold	threshold + shrink
Exact zeros	no	no	yes	yes
Invariant to $X \mapsto XA$	yes	no	no	no
Groups correlated vars	—	yes	no (arbitrary pick)	yes

Table 1: Comparison of OLS, ridge, LASSO, and elastic net.

4.4 Degenerate case: identical predictors

If $x_1 = x_2$, the fit depends on β only through $\beta_1 + \beta_2$, so the RSS contours are straight lines of slope -1 — exactly parallel to the ℓ_1 constraint edge $\beta_1 + \beta_2 = t$. The *entire* edge segment between $(t, 0)$ and $(0, t)$ minimizes the LASSO objective, so the solution is non-unique and numerically unstable. The added ℓ_2 term curves the elastic-net constraint boundary strictly outward, so it touches the linear RSS contour at a single point; by the symmetry of $x_1 = x_2$ this unique tangency point is $\widehat{\beta}_1 = \widehat{\beta}_2 = t/2$.

4.5 Practical guidance

- **OLS**: $n \gg p$, low collinearity, classical inference (unbiasedness, exact t/F-tests) required.
- **Ridge**: collinear predictors or $p \gtrsim n$; belief that most predictors contribute a small effect; pure prediction goal, selection not required.
- **LASSO**: sparse true model believed; interpretability/parsimony desired; predictors roughly uncorrelated.
- **Elastic Net**: many correlated predictors (e.g. genomics, spectra), $p > n$ with possibly $> n$ true active features, need both sparsity and stable grouping.

4.6 Extensions

- **Adaptive LASSO** (Zou, 2006): weight each penalty by $w_j = 1/|\widehat{\beta}_j^{\text{init}}|^\gamma$ from an initial consistent estimate; large true coefficients are penalized less, small ones more, restoring the *oracle property* (recovers the true active set w.p. $\to 1$, with efficient asymptotic normality on the active coefficients).
- **Group LASSO** (Yuan & Lin, 2006): penalize $\sum_g \sqrt{p_g}\|\beta_g\|_2$ over predefined groups g (e.g. dummy blocks of a categorical variable), zeroing out entire groups simultaneously.
- **ℓ_0 penalty**: the "ideal" subset-selection penalty $\lambda\|\beta\|_0$ yields *hard*-thresholding $\widehat{\beta}_j^{\text{hard}} = b_j \mathbb{1}\{|b_j| > \sqrt{\lambda}\}$ in the orthonormal case, but is discontinuous (high variance) and NP-hard to compute in general ($\binom{p}{k}$ search); ℓ_1 is its tightest convex relaxation.
- **Effective degrees of freedom**: for ridge, $\mathrm{df}(\lambda) = \mathrm{tr}(H_\lambda) = \sum_j d_j^2/(d_j^2 + \lambda)$ as above; for LASSO, Zou, Hastie & Tibshirani (2007) show the (Stein-unbiased) effective df equals the number of active (non-zero) coefficients, $\mathrm{df}(\widehat{\beta}(\lambda)) = \#\{j : \widehat{\beta}_j(\lambda) \neq 0\}$.

5 Regression Splines

5.1 Motivation

A single global polynomial is often too rigid: increasing its degree to fit local curvature in one region distorts the fit everywhere else (Runge-type behaviour). Instead, partition the range of x using knots $t_1 < t_2 < \cdots < t_k$ and fit a function that is a low-degree polynomial on each subinterval and joins continuously (and smoothly, for higher-order splines) at every knot.

5.2 Piecewise-linear basis

Define the truncated-linear ("positive part") basis functions $(x - t)_+ = \max(x - t, 0)$. Then

$$\mathcal{B} = \{1,\, x,\, (x - t_1)_+, \ldots, (x - t_k)_+\}$$

spans exactly the space of functions that are linear on each (t_{i-1}, t_i) and continuous at every knot: any such f can be written

$$f(x) = \alpha_0 + \alpha_1 x + \alpha_2 (x - t_1)_+ + \cdots + \alpha_{k+1}(x - t_k)_+.$$

Proof

Why this basis works. Continuity is automatic because $(x - t_i)_+ = 0$ for $x \le t_i$, so adding this term introduces no jump in value at t_i — only a change in slope, from whatever it was to (slope)+α_{i+1}, for $x > t_i$. Conversely, given any continuous piecewise-linear f with slopes $m_0, m_1, \ldots, m_k$ on the $k+1$ pieces, set $\alpha_1 = m_0$ and $\alpha_{i+1} = m_i - m_{i-1}$ for $i = 1, \ldots, k$; then $f(x) = f(t_0^-) + \alpha_1 x + \sum_i \alpha_{i+1}(x - t_i)_+$ reproduces the correct slope on every piece by construction, and α_0 absorbs the additive constant. Hence every such f is a linear combination of $\mathcal{B}$, i.e. $\mathcal{B}$ spans the space; linear independence of $\mathcal{B}$'s $k+2$ functions is immediate from evaluating f and its successive slope-changes at $t_1 < \cdots < t_k$.

Once the basis is fixed, fitting is *ordinary linear regression* on the new "design matrix" of basis-function evaluations:

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} 1 & x_1 & (x_1 - t_1)_+ & \cdots & (x_1 - t_k)_+ \\ \vdots & & & & \\ 1 & x_n & (x_n - t_1)_+ & \cdots & (x_n - t_k)_+ \end{pmatrix}, \quad \widehat{\beta} = (X'X)^{-1}X'Y.$$

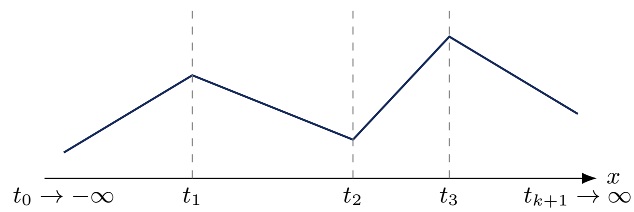

Figure 3: A piecewise-linear spline: linear on each subinterval, continuous (kinked) at each knot.

5.3 Cubic splines and general smoothness order

For cubic splines, the truncated-power basis is

$$\{1,\, x,\, x^2,\, (x - t_1)_+^3, \ldots, (x - t_k)_+^3\},$$

which spans functions that are cubic on each piece and *continuous along with their first and second derivatives* at every knot (visually smooth, no visible kink). More generally $\{1, x, \ldots, x^d, (x - t_i)_+^d\}$ gives degree-d splines with continuity of derivatives up to order $d - 1$ at the knots; $d = 1, 2$ give piecewise linear/quadratic splines respectively.

Proof

Smoothness at a knot. Consider local quadratic representations on either side of t_i: $\alpha_0 + \beta_1 x + \beta_2 x^2$ for $x < t_i$ and $\alpha_0 + \beta_1' x + \beta_2' x^2$ for $x > t_i$ (using the truncated-power basis, the two pieces automatically share $\alpha_0, \beta_1, \beta_2$ up through the added term). Matching value at $x = t_i$ requires $\alpha_0 + \beta_1 t_i + \beta_2 t_i^2 = \alpha_0 + \beta_1' t_i + \beta_2' t_i^2$; matching the first derivative requires $\beta_1 + 2\beta_2 t_i = \beta_1' + 2\beta_2' t_i$. Because $(x - t_i)_+^d$ and its first $d - 1$ derivatives vanish at $x = t_i$ from both sides (they are identically 0 for $x \le t_i$, and the function together with derivatives up to order $d - 1$ is continuous there from the right by direct differentiation), these matching conditions are satisfied automatically by construction whenever the piecewise polynomial is expressed in the truncated power basis of degree d.

Degrees of freedom. A piecewise-linear spline with k interior knots has basis size $k + 2$. More generally, a naive count of "$(k + 1)$ separate cubics, each with 4 free coefficients" gives $4(k + 1)$ parameters; imposing continuity of the function, first, and second derivative at each of the k knots removes $3k$ degrees of freedom, leaving $4(k + 1) - 3k = k + 4$. (Some treatments count only the two lower-order continuity constraints beyond the shared leading terms, giving the frequently quoted $3(k + 1) - 2k = k + 3$ — the exact count depends on precisely how the polynomial pieces are parametrized; using the truncated-power basis directly, the basis $\{1, x, x^2, x^3, (x - t_1)_+^3, \ldots, (x - t_k)_+^3\}$ has exactly $k + 4$ elements, which is the cleanest way to see the correct count.)

B-splines. The truncated power basis is simple but numerically ill-conditioned for many knots (the basis functions are highly correlated / non-local). The **B-spline basis** is an equivalent basis for the same function space whose elements have *compact support* (each nonzero over only a few adjacent knot intervals), giving a banded, well-conditioned design matrix — the standard choice in software.

5.4 Choosing the number and placement of knots

- **Number**: start with many knots (closely, evenly spaced), then apply *backward deletion* — repeatedly remove the least useful knot — to obtain a nested sequence of models, and select one via a criterion such as adjusted R^2, exactly analogous to subset selection (§1.2).

- **Placement**: equally spaced knots waste resolution where f is flat and under-resolve where f curves quickly. Better to place knots at **order statistics / quantiles** of the observed x_i's, so knot density adapts to where data are concentrated (e.g. one knot per equal-count bin of the data) rather than being fixed a priori in x-space.

5.5 Multiple predictors: additive models

With predictors X_1, X_2, lay a grid of knots over each axis and form the combined basis

$$\{1,\, x_1,\, (x_1 - t_{11})_+, \dots,\, x_2,\, (x_2 - t_{21})_+, \dots\},$$

which corresponds to assuming the **additive model**

$$f(x_1, x_2) = \alpha_0 + \underbrace{\alpha_1 x_1 + \cdots + \alpha_{k+1}(x_1 - t_{1k})_+}_{f_1(x_1)} + \underbrace{\beta_1 x_2 + \cdots + \beta_{k+1}(x_2 - t_{2k})_+}_{f_2(x_2)} = f_1(x_1) + f_2(x_2).$$

This extends naturally to p predictors as $f(\boldsymbol{x}) = \sum_{j=1}^{p} f_j(x_j)$ (a **generalized additive model**, fit e.g. by *backfitting*: cycle through predictors, regressing the partial residual $Y - \sum_{l \neq j} \hat{f}_l(x_l)$ nonparametrically on x_j).

5.6 Limitation of additivity and Projection Pursuit Regression

An additive model cannot represent a pure interaction such as $f(x_1, x_2) = x_1 x_2$, since no choice of univariate f_1, f_2 satisfies $x_1 x_2 = f_1(x_1) + f_2(x_2)$ (differentiate twice: $\partial^2(f_1 + f_2)/\partial x_1 \partial x_2 \equiv 0 \neq 1 = \partial^2(x_1 x_2)/\partial x_1 \partial x_2$). However $x_1 x_2$ *can* be written as a sum of ridge functions along *rotated* directions:

$$x_1 x_2 = \left(\frac{x_1 + x_2}{2}\right)^2 - \left(\frac{x_1 - x_2}{2}\right)^2 = f_1(\alpha_1' \boldsymbol{x}) + f_2(\alpha_2' \boldsymbol{x}),$$

with $\alpha_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $\alpha_2 = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, $f_1(t) = t^2/2$, $f_2(t) = -t^2/2$ (direct algebra: $f_1(\alpha_1'\boldsymbol{x}) + f_2(\alpha_2'\boldsymbol{x}) = \frac{1}{2}\big(\frac{x_1+x_2}{\sqrt{2}}\big)^2 - \frac{1}{2}\big(\frac{x_1-x_2}{\sqrt{2}}\big)^2 = \frac{1}{4}(x_1+x_2)^2 - \frac{1}{4}(x_1-x_2)^2 = x_1 x_2$). This motivates the general **Projection Pursuit Regression (PPR)** model

$$f(x_1, \dots, x_p) = \sum_{k=1}^{M} f_k(\alpha_k' \boldsymbol{x}), \qquad \alpha_k \in \mathbb{R}^p,\ M \in \mathbb{N},$$

each term a univariate *ridge function* applied to a linear projection $\alpha_k' \boldsymbol{x}$ of the input; taking each α_k to be a coordinate vector recovers the additive model as a special case, while general α_k allow arbitrary projection directions. PPR generalizes additivity exactly as needed to capture interaction terms like $x_1 x_2$.

Part II: Neural Networks and Nonparametric Regression

6 From Linear Models to Neural Networks

We observe $(X_1, Y_1), \dots, (X_n, Y_n)$, $X_i \in \mathbb{R}^p$, and wish to estimate $f(x) = \mathbb{E}[Y \mid X = x]$. The classical linear model $f(x) = w_0 + \sum_i w_i x_i$ can be generalized in two directions simultaneously explored in projection-pursuit and neural-network models: (i) pass the linear index through a fixed nonlinear *activation/link* function; (ii) *compose* several such transforms (hidden layers), or *sum* several ridge functions (projection pursuit / additive models).

6.1 The single-layer perceptron

Single-layer perceptron

Given inputs $X_0 \equiv 1, X_1, \dots, X_p$ and weights $w_0, \dots, w_p$,

$$Y = \phi\Big(w_0 + \sum_{i=1}^{p} w_i X_i\Big),$$

where $\phi : \mathbb{R} \to \mathbb{R}$ is a fixed, known, typically monotone *transfer (activation) function.*

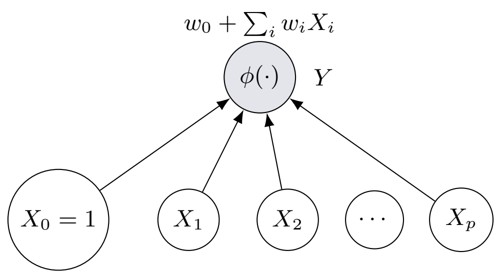

Figure 4: Single-layer perceptron: a weighted sum of inputs passed through a fixed transfer function.

Special cases of ϕ.

- $\phi(t) = t$ (identity) $\Rightarrow$ ordinary linear regression.
- **Sigmoid/logistic**: $\phi(t) = 1/(1 + e^{-t}) \in (0, 1)$, the canonical choice for classification (logistic regression is exactly this model applied to a binary response) — a *generalized linear model* with known link ϕ^{-1}.
- **ReLU**: $\phi(t) = \max(0, t)$, the default in modern deep networks.

Classification rule. With ϕ the sigmoid, $Y = \phi(w_0 + \sum w_i X_i) \in (0, 1)$ is interpreted as $\widehat{P}(Y = 1 \mid X)$; classifying by thresholding Y against a cutoff c is equivalent (since ϕ is strictly increasing) to thresholding the *linear predictor* directly: there is $k_0 = \phi^{-1}(c)$ such that $w_0 + \sum_i w_i X_i \gtrless k_0$ determines the class. The decision boundary $w_0 + \sum_i w_i X_i = k_0$ is therefore a **hyperplane** — the perceptron is fundamentally a linear classifier in the augmented feature space, exactly as in LDA-type methods (§17).

6.2 The multilayer perceptron (MLP)

A single linear index, however transformed, can only capture a restricted (single-ridge/GLM-type) dependence. Introduce k **hidden units**, each itself a perceptron:

$$H_j = f_j\Big(w_{0j} + \sum_{i=1}^{p} w_{ij} X_i\Big), \quad j = 1, \dots, k, \qquad Y = f\Big(w_0 + \sum_{j=1}^{k} w_j H_j\Big).$$

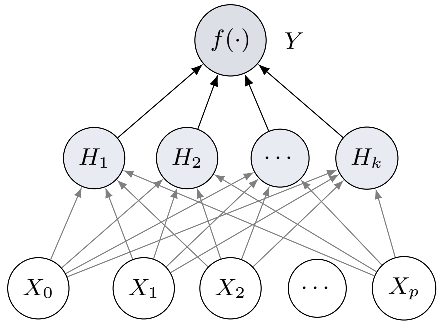

Figure 5: Multilayer perceptron with one hidden layer: each H_j is itself a perceptron applied to X.

If the output activation is linear, $f(t) = t$: writing $\alpha_j = (w_{0j}, \ldots, w_{pj})'$ and absorbing w_0,

$$Y = \sum_{j=1}^{k} w_j f_j(\alpha_j' X) = \sum_{j=1}^{k} f_j^*(\alpha_j' X),$$

which is exactly the **projection pursuit** form (§7) — the MLP with linear output and one hidden layer *is* a projection-pursuit model in which every ridge function f_j is forced to be the *same known* sigmoid, whereas PPR estimates each f_j freely and nonparametrically.

6.3 The Universal Approximation Theorem

Theorem 6.1 (Hornik–Stinchcombe–White, 1989)**.** *Let* $f_{\mathrm{MLP}}(x) = w_0 + \sum_{j=1}^{k} w_j \sigma(\alpha_j' x)$*,* σ *bounded, non-constant, monotone (sigmoidal). Then for every continuous* f *on a compact* $\mathcal{X} \subset \mathbb{R}^p$ *and every* $\varepsilon > 0$*, there exist* $k \in \mathbb{N}$ *and parameters such that* $\sup_{x \in \mathcal{X}} |f(x) - f_{\mathrm{MLP}}(x)| < \varepsilon$*.*

Proof

Proof sketch. (i) Let $\Sigma_p = \mathrm{span}\{\sigma(\alpha' x + b) : \alpha \in \mathbb{R}^p, b \in \mathbb{R}\}$, a linear subspace of $C(\mathcal{X})$ under the sup-norm. (ii) Suppose, for contradiction, that $\overline{\Sigma_p} \subsetneq C(\mathcal{X})$. By Hahn–Banach there is a nonzero bounded linear functional L vanishing on Σ_p; by Riesz representation, $L(g) = \int_{\mathcal{X}} g \, d\mu$ for a finite signed regular Borel measure μ. (iii) Vanishing on all ridge functions gives $\int \sigma(\alpha' x + b) \, d\mu(x) = 0$ for all α, b. (iv) Taking $b \to \pm\infty$ (appropriately rescaled) and using monotonicity/boundedness of σ, this forces the Fourier–Stieltjes transform of μ, restricted to hyperplanes, to vanish identically; by uniqueness of such transforms, $\mu \equiv 0$ — contradicting $L \neq 0$. Hence $\overline{\Sigma_p} = C(\mathcal{X})$: finite sums of sigmoidal ridge functions are dense in $C(\mathcal{X})$.

This is a pure *existence* result — it guarantees some k and weights exist, but says nothing about how large k must be for a given accuracy or how to estimate the weights from finite data n (a separate statistical problem addressed by nonlinear least squares / backpropagation, where the usual bias–variance trade-off between k and estimation variance applies).

7 Projection Pursuit Regression

7.1 Model and comparison with the MLP

$$f(x_1, \ldots, x_p) = \sum_{j=1}^{k} f_j(\alpha_j' x), \qquad \|\alpha_j\| = 1,$$

with *both* the directions $\{\alpha_j\}$ and the ridge functions $\{f_j\}$ estimated from data (unlike the MLP, where f_j is a fixed known sigmoid).

7.2 Estimation by backfitting

1. Fix all terms but the m-th; form the partial residual $R_i^{(m)} = Y_i - \sum_{j \neq m} f_j(\alpha_j' X_i)$. 2. Update f_m given α_m: a univariate scatterplot-smoothing problem — regress $R_i^{(m)}$ nonparametrically on the scalars $\alpha_m' X_i$ (kernel/local smoother, spline).

	MLP (1 hidden layer, linear output)	Projection Pursuit Regression
Ridge functions f_j	fixed, known (sigmoid)	unknown, estimated nonparametrically
Estimated	weights w_j, α_j (finite-dim.)	directions α_j *and* functions f_j
Flexibility source	many hidden units k	flexible f_j's with few terms k

3. Update α_m given f_m: minimize $\sum_i (R_i^{(m)} - f_m(\alpha' X_i))^2$ over α (a Gauss–Newton step), subject to $\|\alpha_m\| = 1$. 4. Cycle over $m = 1, \ldots, k$ until convergence.

8 Local Averaging: k-Nearest Neighbor Regression

Moving from parametric/semi-parametric models to fully nonparametric *local averaging*: estimate $f(x)$ by averaging the Y_i's whose X_i's are near x, with no assumed functional form.

8.1 The k-NN estimator

Let $N_k(x)$ index the k points among $X_1, \ldots, X_n$ closest to x. Then

$$\hat{f}_k(x) = \frac{1}{k} \sum_{i: X_i \in N_k(x)} Y_i.$$

Theorem 8.1 (Consistency conditions)**.** *For $\hat{f}_k(x) \to f(x)$ as $n \to \infty$ (under standard smoothness conditions), it is necessary that*

$$k \to \infty \qquad \textit{and} \qquad k/n \to 0.$$

Proof

Decompose $\mathbb{E}[(\hat{f}_k(x) - f(x))^2] = (\mathbb{E}[\hat{f}_k(x)] - f(x))^2 + \mathrm{Var}(\hat{f}_k(x))$. **Variance:** writing $\hat{f}_k(x) = \frac{1}{k} \sum_{i \in N_k(x)} (f(X_i) + \varepsilon_i)$ with $\mathrm{Var}(\varepsilon_i) = \sigma^2$ and treating the k neighbor noise terms as approximately uncorrelated, $\mathrm{Var}(\hat{f}_k(x)) \approx \sigma^2/k$. If k stayed bounded, this would not vanish, so $k \to \infty$ is needed. **Bias:** the k nearest neighbors occupy a ball of radius $r_k(x)$ around x; by properties of order statistics of the covariate distribution, $r_k(x) \to 0$ iff $k/n \to 0$. If instead $k = cn$ for fixed $c > 0$, $N_k(x)$ continues to cover a non-shrinking region even as $n \to \infty$, so $\mathbb{E}[\hat{f}_k(x)] \approx \frac{1}{k} \sum_{i \in N_k(x)} f(X_i) \not\to f(x)$ unless f happens to be locally constant there. Hence $k/n \to 0$ is needed for the bias to vanish. Both are needed simultaneously for MSE $\to 0$.

8.2 Weighted k-NN and its discontinuity

$$\hat{f}_{k.WNN}(x) = \frac{\sum_{i \in N_k(x)} \omega(\|x - X_i\|) Y_i}{\sum_{i \in N_k(x)} \omega(\|x - X_i\|)},$$

ω non-increasing so nearer points get larger weight. Even if ω is continuous, $x \mapsto \hat{f}_{k.WNN}(x)$ can be *discontinuous* (unless $k = n$): as x crosses a perpendicular bisector between two data points, the set $N_k(x)$ changes discretely — one neighbor abruptly leaves and another enters — producing a jump.

8.3 Kernel (Nadaraya–Watson) regression

Take $k = n$: use *all* points, letting a continuous weight function alone localize the fit:

$$\hat{f}(x) = \frac{\sum_{i=1}^{n} \omega(\|x - X_i\|) Y_i}{\sum_{i=1}^{n} \omega(\|x - X_i\|)}.$$

Proposition 8.2. *If ω is continuous and strictly positive, $\hat{f}(x)$ is continuous in x.*

Proof

$N(x) = \sum_i \omega(\|x - X_i\|)Y_i$ and $D(x) = \sum_i \omega(\|x - X_i\|)$ are finite sums of compositions of the continuous map $x \mapsto \|x - X_i\|$ with continuous ω, hence continuous; with $D(x) > 0$ everywhere, the ratio $\hat{f} = N/D$ is continuous. Unlike k-NN, no point ever abruptly enters/exits the average — every point contributes at every x with a continuously varying weight.

A popular choice is the **Gaussian weight** $\omega(t) = e^{-t^2/2}$; introducing a **bandwidth** $h > 0$ and kernel $K(t) = e^{-\|t\|^2/2}$,

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-X_i}{h}\right)Y_i}{\sum_{i=1}^n K\left(\frac{x-X_i}{h}\right)} \qquad \textbf{(Nadaraya–Watson estimator)}.$$

Other valid kernels (Epanechnikov, uniform, triangular) may replace K.

Proposition 8.3 (NW as local weighted least squares)**.** $\hat{f}_h(x) = \arg\min_{c\in\mathbb{R}} \sum_i K_h(x-X_i)(Y_i-c)^2$, *i.e. the local constant weighted-LS fit at* x.

Proof

$Q(c) = \sum_i K_h(x - X_i)(Y_i - c)^2$ is convex quadratic in c; $dQ/dc = -2\sum_i K_h(x - X_i)(Y_i - c) = 0 \Rightarrow c^* = \sum_i K_h(x - X_i)Y_i / \sum_i K_h(x - X_i) = \hat{f}_h(x)$.

This exhibits kernel regression as the 0-th order member of the broader **local polynomial regression** family, which fits a local linear/quadratic/... polynomial (instead of a local constant) weighted by $K_h(x - X_i)$ at every x — local linear fits reduce boundary bias relative to Nadaraya–Watson.

8.4 Bandwidth selection

As with λ in ridge/LASSO, h controls a bias–variance trade-off (small h: low bias, high variance; large h: high bias, low variance) and is typically chosen by cross-validation, minimizing $\mathrm{CV}(h) = \sum_i (Y_i - \hat{f}_h^{(-i)}(X_i))^2$ (leave-one-out).

Summary chain of ideas: k-NN (unweighted) $\to$ weighted k-NN (discontinuous unless $k = n$) $\to$ full-sample weighted average ($k = n$) $\to$ Gaussian weight $\to$ add bandwidth h $\to$ general kernel K $\to$ Nadaraya–Watson.

Part III: Trees and Ensembles

9 Regression Trees

9.1 Motivation and formulation

Kernel/k-NN regression produces smooth local averages; **regression trees** instead produce an *adaptively partitioned, piecewise-constant* estimate: the covariate space $\mathbb{R}^p$ is recursively split into disjoint rectangular regions $R_1, \ldots, R_M$ (the leaves) via a sequence of binary splits $\{X_j < s\}$ vs. $\{X_j \geq s\}$, and

$$\hat{f}(x) = \sum_{m=1}^{M} \hat{c}_m \mathbb{1}\{x \in R_m\}, \qquad \hat{c}_m = \frac{1}{n_m} \sum_{i:X_i \in R_m} Y_i.$$

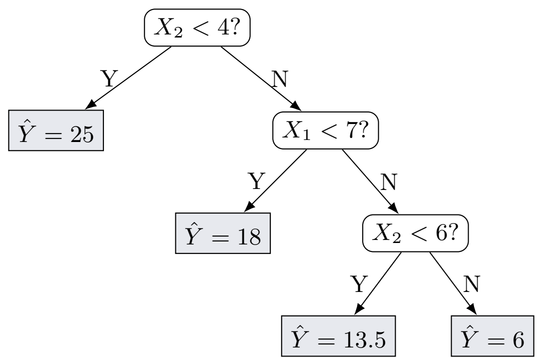

Figure 6: An illustrative regression tree: each internal node asks a yes/no question about one covariate.

9.2 Node impurity and the splitting criterion

For a node t with n observations $y_1, \ldots, y_n$, define **impurity** as the within-node variance

$$i(t) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \bar{y}_t)^2, \qquad \bar{y}_t = \frac{1}{n} \sum_i y_i.$$

A split divides t into t_L (n_L obs.) and t_R (n_R obs.), $n_L + n_R = n$. We choose the split (j^*, s^*) minimizing the weighted child impurity, equivalently maximizing the impurity reduction

$$\Delta i(t, s) = i(t) - \Big[\tfrac{n_L}{n} i(t_L) + \tfrac{n_R}{n} i(t_R)\Big].$$

Proposition 9.1 (Variance decomposition across a split)**.**

$$n\, i(t) = n_L i(t_L) + n_R i(t_R) + \frac{n_L n_R}{n} (\bar{y}_{t_L} - \bar{y}_{t_R})^2, \qquad \Delta i(t, s) = \frac{n_L n_R}{n^2} (\bar{y}_{t_L} - \bar{y}_{t_R})^2 \geq 0.$$

Proof

Write $ni(t) = \sum_{i \in t} (y_i - \bar{y}_t)^2$ and split the sum over children. For $i \in t_L$, using $y_i - \bar{y}_t = (y_i - \bar{y}_{t_L}) + (\bar{y}_{t_L} - \bar{y}_t)$,

$$\sum_{i \in t_L} (y_i - \bar{y}_t)^2 = \sum_{i \in t_L} (y_i - \bar{y}_{t_L})^2 + 2(\bar{y}_{t_L} - \bar{y}_t) \sum_{i \in t_L} (y_i - \bar{y}_{t_L}) + n_L (\bar{y}_{t_L} - \bar{y}_t)^2.$$

The cross term vanishes ($\sum_{i \in t_L} (y_i - \bar{y}_{t_L}) = 0$), leaving $n_L i(t_L) + n_L (\bar{y}_{t_L} - \bar{y}_t)^2$, and symmetrically for t_R. Summing:

$$ni(t) = n_L i(t_L) + n_R i(t_R) + \big[n_L (\bar{y}_{t_L} - \bar{y}_t)^2 + n_R (\bar{y}_{t_R} - \bar{y}_t)^2\big].$$

Since $\bar{y}_t = (n_L\bar{y}_{t_L} + n_R\bar{y}_{t_R})/n$, we get $\bar{y}_{t_L} - \bar{y}_t = \frac{n_R}{n}(\bar{y}_{t_L} - \bar{y}_{t_R})$ and $\bar{y}_{t_R} - \bar{y}_t = -\frac{n_L}{n}(\bar{y}_{t_L} - \bar{y}_{t_R})$. Substituting and using $n_L + n_R = n$,

$$n_L(\bar{y}_{t_L} - \bar{y}_t)^2 + n_R(\bar{y}_{t_R} - \bar{y}_t)^2 = \frac{n_L n_R(n_R + n_L)}{n^2}(\bar{y}_{t_L} - \bar{y}_{t_R})^2 = \frac{n_L n_R}{n}(\bar{y}_{t_L} - \bar{y}_{t_R})^2.$$

Dividing by n and rearranging gives $\Delta i(t, s)$ as claimed; it is manifestly ≥ 0, zero iff $\bar{y}_{t_L} = \bar{y}_{t_R}$.

This is exactly the between/within sums-of-squares decomposition of one-way ANOVA: minimizing weighted child impurity is equivalent to **maximizing the (weighted) squared separation of the two children's means**, a computationally convenient purely mean-based split criterion.

9.3 Growing algorithm

1. Start with the root containing all n observations.
2. At each current leaf t, and for every covariate X_j and candidate split s (typically midpoints of consecutive order statistics of X_j within t), compute $\Delta i(t, s)$.
3. Choose $(j^*, s^*) = \arg\max \Delta i(t, s)$; split t accordingly.
4. Recurse on each child.
5. Stop by a stopping rule (below); assign each leaf's fitted value as its mean response.

Stopping criteria: minimum node size (minsplit/minbucket); impurity-reduction threshold; maximum depth; or — generally preferred — grow a large tree first and *prune* (below), since greedy one-step stopping can miss a weak split that enables a very informative split in its child (non-monotone lookahead problem).

10 Minimal Cost-Complexity Pruning (BFOS, 1984)

10.1 Global tree impurity and the cost-complexity criterion

For a tree T with leaves T_e, define global impurity $i(T) = \sum_{t \in T_e} p_t i(t)$, $p_t = n_t/n$. Breiman, Friedman, Olshen & Stone (1984) define, for any subtree $T \subseteq T_0$,

$$R_\alpha(T) = R(T) + \alpha|T_e|, \qquad R(T) = \sum_{t \in T_e} \sum_{x_i \in t} (y_i - \hat{y}_t)^2 \text{ (regression RSS)},$$

with $\alpha \geq 0$ a complexity penalty per leaf.

10.2 Derivation of the break-point sequence

At $\alpha = 0$, $R_0(T_M) \leq R_0(T)$ for all $T \subseteq T_M$ (splitting never increases training RSS), so the full tree T_M is optimal. For $\alpha > 0$, a smaller tree T_k ($k < M$ leaves) beats T_M iff

$$R_\alpha(T_k) < R_\alpha(T_M) \iff R(T_k) + \alpha k < R(T_M) + \alpha M \iff \alpha > \frac{R(T_k) - R(T_M)}{M - k}$$

(using $M - k > 0$). The critical transition value is the *minimum* such slope over all candidates,

$$\alpha_0 = \min_{1 \leq k < M} \frac{R(T_k) - R(T_M)}{M - k}.$$

Iterating this comparison across successively smaller trees yields a strictly increasing sequence $0 = \alpha_0 < \alpha_1 < \cdots < \alpha_K < \infty$ partitioning $[0, \infty)$ into intervals $[\alpha_{j-1}, \alpha_j)$ on each of which $\arg\min_T R_\alpha(T)$ is the same unique subtree T_{M_j}, giving a nested sequence $T_M \succ T_{M_1} \succ \cdots \succ T_1$ (the root).

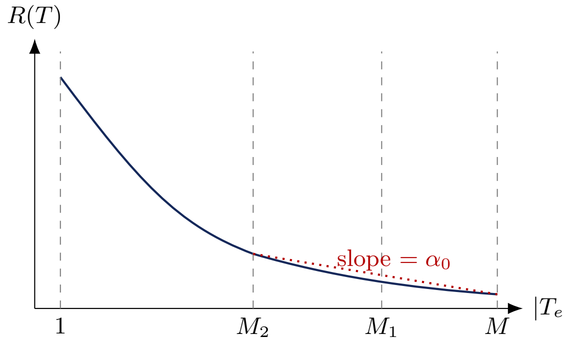

Figure 7: Geometric picture of α_0 as the minimal slope of $R(T)$ vs. $|T_e|$ over candidate leaf reductions.

Automatic pruning of intermediate sizes

Cost-complexity pruning need not produce trees of every size $M, M-1, M-2, \dots$: if collapsing a whole multi-level branch (several splits at once) gives a better drop in R_α than any single-leaf merge, those intermediate sizes are skipped automatically. This is why cost-complexity pruning strictly dominates the naive greedy bottom-up leaf-merging heuristic (repeatedly collapsing the locally cheapest sibling pair), which is myopic and can miss globally superior branch collapses.

10.3 Selecting α by cross-validation

Since every α in $[\alpha_{j-1}, \alpha_j)$ gives the same optimal subtree, pick a representative $\alpha_j^* = \sqrt{\alpha_{j-1}\alpha_j}$ (geometric mean) or $(\alpha_{j-1} + \alpha_j)/2$ (arithmetic mean) per interval, then:

1. Partition the data into V folds $D_1, \dots, D_V$.
2. For each fold v: grow a full tree on $D_{(-v)} = D \setminus D_v$; for each α_j^*, prune to $T^{(-v)}(\alpha_j^*)$; compute $PE_v(\alpha_j^*) = \sum_{i \in D_v} (y_i - \hat{f}^{(-v)}(x_i; \alpha_j^*))^2$.
3. $\mathrm{CV}(\alpha_j^*) = \sum_v PE_v(\alpha_j^*)$; choose $\hat{\alpha} = \arg\min \mathrm{CV}(\alpha_j^*)$.
4. Return $T^* = T_0(\hat{\alpha})$, pruning the *original* full tree by $\hat{\alpha}$.

11 Classification Trees and Impurity Axiomatics

11.1 Binary classification via regression variance

Coding $Y \in \{0, 1\}$ (class B/A), with n_A, n_B counts in a node, $p = n_A/n = \bar{y}_t$: since $y_i^2 = y_i$ for binary y_i, the regression-variance impurity reduces to

$$i(t) = \frac{1}{n} \sum y_i^2 - \bar{y}_t^2 = p - p^2 = p(1-p) = p_A p_B = n_A n_B / n^2,$$

which is exactly the **binary Gini index**. The Bayes-optimal (majority-vote) leaf prediction is $\hat{Y}(x) = 1$ if $\bar{y}_t > 0.5$, else 0.

11.2 Multi-class impurity axioms

For $K \geq 2$ classes with empirical vector $p = (p_1, \dots, p_K) \in \Delta_{K-1}$ (the probability simplex), impurity is $i(t) = \Psi(p)$ for a scalar $\Psi : \Delta_{K-1} \to \mathbb{R}$. A valid impurity measure should satisfy:

1. **Symmetry**: $\Psi(p_1, \dots, p_K) = \Psi(p_{\pi(1)}, \dots, p_{\pi(K)})$ for any permutation π.
2. **Maximality at uniform**: $\arg\max_p \Psi(p) = (1/K, \dots, 1/K)$.
3. **Minimality at purity**: Ψ is minimized iff $p_j = 1$ for some j.

4. **Non-negative impurity reduction**: every split satisfies $\Delta i(t) = \Psi(p) - \left[\frac{n_L}{n_t}\Psi(p^L) + \frac{n_R}{n_t}\Psi(p^R)\right] \geq 0$.

Sufficiency theorem

If Ψ is symmetric and *concave* on Δ_{K-1}, it satisfies Properties 1–4.

Proof

Property 2 (maximum at uniform). For $p \in \Delta_{K-1}$, consider the K cyclic permutations $p^{(j)}$. By symmetry $\Psi(p^{(j)}) = \Psi(p) \equiv \Psi_0$ for all j. Their average is $\bar{p} = \frac{1}{K}\sum_j p^{(j)}$, and computing the r-th coordinate, $\bar{p}_r = \frac{1}{K}\sum_k p_k = 1/K$ for every r, i.e. $\bar{p} = (1/K, \ldots, 1/K)$. By concavity (Jensen), $\Psi(\bar{p}) \geq \frac{1}{K}\sum_j \Psi(p^{(j)}) = \Psi(p)$, so $\Psi(1/K, \ldots, 1/K) \geq \Psi(p)$ for all p.
Property 3 (minimum at purity). Let e_k be the pure-class vectors; $\Psi(e_k) \equiv \Psi^*$ for all k by symmetry. Any $p = \sum_k p_k e_k$ (convex combination), so by concavity $\Psi(p) \geq \sum_k p_k \Psi(e_k) = \Psi^* \sum_k p_k = \Psi^*$.
Property 4 (non-negative reduction). Class counts satisfy $n_{t,k} = n_{L,k} + n_{R,k}$, so $p = \frac{n_L}{n}p^L + \frac{n_R}{n}p^R$ is a convex combination; by concavity $\Psi(p) \geq \frac{n_L}{n}\Psi(p^L) + \frac{n_R}{n}\Psi(p^R)$, i.e. $\Delta i(t) \geq 0$.

11.3 Canonical impurity criteria

Metric	Formula	Value at uniform	Value at pure
Gini index	$1 - \sum_k p_k^2 = \sum_{i \neq j} p_i p_j$	$\frac{K-1}{K}$	0
Cross-entropy	$-\sum_k p_k \log_2 p_k$	$\log_2 K$	0
Misclassification	$1 - \max_k p_k$	$\frac{K-1}{K}$	0

Proof

Gini is strictly concave. Using $\sum_{i \neq j} p_i p_j = 1 - \sum_k p_k^2$ (expand $(\sum_k p_k)^2 = \sum_k p_k^2 + \sum_{i \neq j} p_i p_j = 1$), the Hessian of $\Psi_{\text{Gini}}(p) = 1 - \sum_k p_k^2$ is $\nabla^2 \Psi_{\text{Gini}} = -2I_K \prec 0$: strictly negative definite, hence strictly concave.
Cross-entropy is strictly concave. $g(x) = -x \log x$ on $(0,1)$ has $g''(x) = -1/x < 0$; a sum of strictly concave univariate functions $\Psi_{\text{Entropy}}(p) = \sum_k g(p_k)$ is strictly concave.
Misclassification is concave (not strictly). $\max_k p_k = \max_k \{e_k' p\}$ is a pointwise supremum of linear functions, hence convex; $\Psi_{\text{Misclass}}(p) = 1 - \max_k p_k$ is therefore concave (but only piecewise linear, so not strictly — this is why Gini/entropy are preferred in practice: they are strictly concave and more sensitive to changes in class purity, giving finer-grained split comparisons).

12 Bootstrap Aggregation (Bagging)

12.1 Motivation

Deep (unpruned) trees have low bias but very high variance: because splits are greedy and hierarchical, a small perturbation of the training data can alter an early split and cascade into a completely different tree. **Bagging** (Breiman, 1996) stabilizes this by averaging trees fit to many bootstrap resamples.

12.2 Bootstrap sampling theory

Proposition 12.1. *Drawing n points with replacement (SRSWR) from a dataset of size n, the expected fraction of* distinct *original observations appearing in the resample converges to $1 - e^{-1} \approx 0.632$.*

Proof

Fix A_i. $P(A_i \text{ not drawn in one pick}) = 1 - 1/n$; over n independent draws, $P(A_i \notin D^*) = (1 - 1/n)^n$, so $P(A_i \in D^*) = 1 - (1 - 1/n)^n$. Let $I_i = \mathbb{1}\{A_i \in D^*\}$, $N_0 = \sum_i I_i$. Then $\mathbb{E}[N_0/n] = 1 - (1 - 1/n)^n \to 1 - e^{-1} \approx 0.632$ as

$n \to \infty$, using $\lim_{n\to\infty}(1 - 1/n)^n = e^{-1}$.

The remaining $\approx 36.8\%$ are **Out-Of-Bag (OOB)** for that bootstrap tree, and provide a built-in validation set for estimating generalization error without extra cross-validation.

12.3 The bagging procedure

1. For $b = 1, \dots, B$: draw $D^{(b)}$ (SRSWR, size n) from D; fit an *unpruned* tree $\hat{f}_b$ on $D^{(b)}$.
2. Aggregate: regression $\hat{f}_{\text{bag}}(x) = \frac{1}{B}\sum_b \hat{f}_b(x)$; classification, majority vote $\hat{C}_{\text{bag}}(x) = \arg\max_k \sum_b \mathbb{1}\{\hat{C}_b(x) = k\}$.

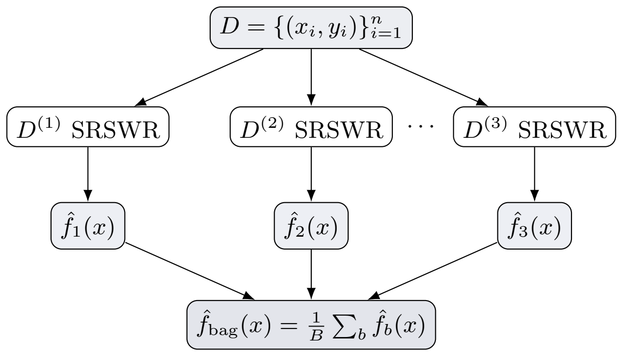

Figure 8: Bootstrap Aggregation: resample, fit unpruned trees, average.

12.4 Variance reduction for correlated ensembles

Proposition 12.2. *Let $Z_1, \dots, Z_m$ be identically distributed,* $\mathrm{Var}(Z_i) = \sigma^2$*, pairwise* $\mathrm{Corr}(Z_i, Z_j) = \rho$ *($i \neq j$). Then* $\mathrm{Var}(\bar{Z}) = \rho\sigma^2 + \frac{1-\rho}{m}\sigma^2$.

Proof

$\mathrm{Var}(\bar{Z}) = \frac{1}{m^2}\sum_i\sum_j \mathrm{Cov}(Z_i, Z_j) = \frac{1}{m^2}\left[m\sigma^2 + m(m-1)\rho\sigma^2\right] = \frac{\sigma^2}{m} + \left(1 - \frac{1}{m}\right)\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{m}\sigma^2.$

As $m \to \infty$, $\mathrm{Var}(\bar{Z}) \to \rho\sigma^2$: increasing the number of bootstrap trees B drives the second term to zero but *cannot* eliminate the **correlation floor** $\rho\sigma^2$. Since all bagged trees are fit on resamples of the *same* data D, $\rho > 0$ in general (dominant predictors tend to appear at the top of most trees), capping bagging's variance-reduction benefit — the motivation for Random Forests.

13 Random Forests

13.1 Feature subsampling

Breiman (2001): in addition to bootstrap resampling, at *each split* restrict the search to a random subset of $p' \ll p$ features (drawn without replacement from the p available), rather than searching all p. Typical defaults: $p' \approx \sqrt{p}$ (classification), $p' \approx p/3$ (regression). This prevents a small set of dominant predictors from being selected at the top of every tree, **decorrelating** the ensemble and lowering ρ (hence the variance floor $\rho\sigma^2$ above) beyond what bagging alone achieves.

Proposition 13.1 (Probability a given feature is used)**.** *With p' of p features sampled per split, the probability a fixed feature X_1 is chosen at a given split is p'/p, and the probability it is selected at least once across m independent splits is $1 - (1 - p'/p)^m$.*

Proof

The number of size-p' subsets is $\binom{p}{p'}$; the number excluding X_1 is $\binom{p-1}{p'}$ (choosing all p' from the remaining $p-1$). So

$$P(X_1 \text{ omitted at one split}) = \frac{\binom{p-1}{p'}}{\binom{p}{p'}} = \frac{(p-1)!/[p'!(p-1-p')!]}{p!/[p'!(p-p')!]} = \frac{p-p'}{p} = 1 - \frac{p'}{p}.$$

Assuming independence across m splits, $P(\text{omitted in all } m) = (1-p'/p)^m$, so $P(\text{selected at least once}) = 1-(1-p'/p)^m$.

13.2 Random forest and stratified classification

1. For $b = 1, \dots, B$: draw $D^{(b)}$ (SRSWR); grow an unpruned tree, at each node restricting the split search to a fresh random size-p' subset of features.
2. Aggregate: $\hat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_b \hat{f}^*_b(x)$ (regression) or majority vote $\delta(x) = \arg\max_j \sum_b \mathbb{1}\{\delta_b(x) = j\}$ (classification).

Rare-class problem: under plain SRSWR across a pooled dataset, a low-prevalence class may be entirely absent from some bootstrap resamples, so some base trees cannot predict it. **Stratified bootstrap**: partition into per-class strata $D_1, \dots, D_J$ and resample n_j points independently *within* each stratum, guaranteeing every class is represented in every resample.

Method	Model structure	Advantages	Limitations
Single tree	discontinuous piecewise constant	interpretable, automatic feature selection	high variance, unstable
Bagged trees	average of B bootstrap trees	reduced variance	retains correlation when dominant features exist
Random forest	average of B feature-randomized trees	decorrelates trees, lower variance floor	loses interpretability

14 Multivariate Adaptive Regression Splines (MARS)

14.1 Motivation and hinge functions

Trees produce discontinuous step-function fits $\hat{f}(x) = \sum_{t \in T_e} c_t \mathbb{1}\{x \in R_t\}$, unsuitable for smooth continuous phenomena without growing impractically deep trees. Friedman (1991) introduced **MARS**, combining continuous piecewise-linear *hinge functions* with the adaptive, recursive split-selection idea of trees. For a knot t, define the **reflected pair**

$$(x-t)_+ = \max(0, x-t), \qquad (t-x)_+ = \max(0, t-x),$$

continuous, joining at zero at $x = t$, with opposite unit slopes on either side.

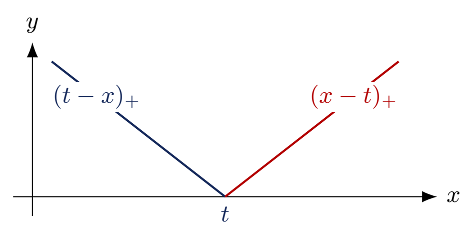

Figure 9: A reflected pair of hinge (truncated-linear) basis functions joining continuously at knot t.

14.2 Univariate MARS

Starting from $B_0(x) = 1$, for each candidate knot $t \in \{x_1, \ldots, x_n\}$: fit $\hat{y}(x) = a_0 + a_1(x - t)_+ + a_2(t - x)_+$ by least squares, giving $RSS(t)$; choose $t^* = \arg\min_t RSS(t)$.

14.3 Multivariate forward/backward passes

Forward stepwise: initialize $\mathcal{M} = \{B_0\}$. At each stage, for every existing basis $B_s \in \mathcal{M}$, predictor X_j, and candidate knot t, form the candidate pair $B_s(x)(X_j - t)_+$, $B_s(x)(t - X_j)_+$, refit by OLS, and record $RSS(s, j, t)$; append the pair (s^*, j^*, t^*) giving the largest RSS reduction. This allows *interaction* terms (products of hinges from different predictors), analogous to a tree's recursive splitting but yielding a continuous surface. Search cost is $O(p \cdot n \cdot m)$ per forward step (with m current basis functions); practitioners typically cap interaction order at 2–3 to control overfitting/noise.

Backward deletion + GCV: the forward pass overfits; remove, one at a time, the basis function whose deletion increases RSS least, producing a nested sequence $\mathcal{M}_{\text{full}} \succ \mathcal{M}_{K-1} \succ \cdots \succ \mathcal{M}_1$, and select the final model via **Generalized Cross-Validation**

$$\text{GCV}(\lambda) = \frac{\frac{1}{n}\sum_i (y_i - \hat{f}_\lambda(x_i))^2}{\left(1 - \tilde{C}(\lambda)/n\right)^2}, \qquad \tilde{C}(\lambda) = r + c \cdot K,$$

r = number of linearly independent basis functions, K = number of forward-pass knots, $c \approx$ 2–3 a per-knot penalty factor — GCV approximates leave-one-out cross-validation without actually refitting n times.

Method	Model structure	Advantages	Limitations
Single tree	discontinuous piecewise constant	interpretable	unstable, discontinuous
MARS	continuous tensor-product linear splines	captures continuous nonlinearity & interactions	forward search is $O(p \cdot n \cdot m)$

Part IV: Statistical Decision Theory and the Bayes Classifier

15 Statistical Decision Theory

15.1 Loss, risk, and decision rules

Let $X \in \mathcal{X} \subseteq \mathbb{R}^d$ be a feature vector and $Y \in \mathcal{Y}$ the target, governed by an unknown joint law $P_{X,Y}$. A **decision rule** is a measurable map $\delta : \mathbb{R}^d \to \mathcal{Y}$; a **loss function** $L(y, \delta(x))$ penalizes discrepancy; the **(theoretical/expected) risk** is $R(\delta) = \mathbb{E}_{X,Y}[L(Y, \delta(X))]$.

15.2 Binary classification under 0–1 loss

For $C \in \{1, 2\}$ with priors $\pi_1 = P(C = 1)$, $\pi_2 = P(C = 2)$ and class-conditional densities f_1, f_2, the 0–1 loss $L(c, \delta(x)) = \mathbb{1}\{c \neq \delta(x)\}$ gives risk $R(\delta) = P(\delta(X) \neq C)$.

Proposition 15.1 (Risk decomposition)**.**

$$R(\delta) = \pi_1 \int_{\{x:\delta(x)=2\}} f_1(x)\, dx + \pi_2 \int_{\{x:\delta(x)=1\}} f_2(x)\, dx.$$

Proof

By the law of total probability, $R(\delta) = P(\delta(X) = 2, C = 1) + P(\delta(X) = 1, C = 2) = P(C{=}1)P(\delta(X){=}2 \mid C{=}1) + P(C{=}2)P(\delta(X){=}1 \mid C{=}2)$, and each conditional probability is the integral of the corresponding density over the (mis-)decision region.

The *empirical* risk $\hat{R}(\delta) = \frac{1}{n}\big[\sum_i I(\delta(x_{1i}) = 2) + \sum_i I(\delta(x_{2i}) = 1)\big]$ is a sum of discontinuous indicator functions: it is non-smooth, non-convex, piecewise-constant, and NP-hard to minimize directly over general classifier families. Two resolutions: (i) derive the exact analytical minimizer of the *population* risk (the Bayes rule, below); (ii) substitute a smooth convex **surrogate loss** (logistic deviance, hinge loss for SVMs, exponential loss for AdaBoost) for empirical optimization.

16 The Bayes Classifier

16.1 Derivation

Bayes rule for two classes

The risk-minimizing decision rule is

$$\delta^*(x) = \begin{cases} 1 & \pi_1 f_1(x) > \pi_2 f_2(x) \\ 2 & \pi_1 f_1(x) < \pi_2 f_2(x) \end{cases}$$

(ties broken arbitrarily without affecting the risk).

Proof

Since $\int_{R_1(\delta)^c} f_1\, dx = 1 - \int_{R_1(\delta)} f_1\, dx$,

$$R(\delta) = \pi_1\Big(1 - \int_{R_1(\delta)} f_1\Big) + \pi_2 \int_{R_1(\delta)} f_2 = \pi_1 + \int_{R_1(\delta)} \big[\pi_2 f_2(x) - \pi_1 f_1(x)\big]\, dx.$$

π_1 does not depend on δ; the choice of whether to include each x in $R_1(\delta)$ can be made independently pointwise, so the integral (hence $R(\delta)$) is minimized by including $x \in R_1(\delta)$ exactly when the integrand is negative, i.e. $\pi_2 f_2(x) < \pi_1 f_1(x)$, and excluding it otherwise.

By Bayes' theorem $P(C=1 \mid X=x) = \pi_1 f_1(x)/[\pi_1 f_1(x) + \pi_2 f_2(x)]$, so $\pi_1 f_1(x) > \pi_2 f_2(x) \iff P(C=1 \mid X=x) > P(C=2 \mid X=x)$: **the Bayes rule assigns x to the class of highest posterior probability**. For K classes this generalizes to $\delta^*(x) = \arg\max_k \pi_k f_k(x) = \arg\max_k P(C=k \mid X=x)$.

16.2 Worked examples

Example 16.1 (Discrete: defective items)**.** Machine 1 produces 60% of items (π_1=0.6), defect rate p_1=0.2; Machine 2 produces 40% (π_2=0.4), p_2=0.3. A pack of $n = 20$ has $X = 5$ defectives. With $X \mid M_k \sim \text{Bin}(20, p_k)$,

$$\frac{\pi_1 f_1(5)}{\pi_2 f_2(5)} = \frac{0.6}{0.4} \cdot \left(\frac{0.2}{0.3}\right)^5 \left(\frac{0.8}{0.7}\right)^{15} = 1.5 \times 0.1317 \times 7.026 \approx 1.388 > 1,$$

so $P(M_1 \mid X=5) \approx 0.581 > P(M_2 \mid X=5) \approx 0.419$: classify as Machine 1.

Example 16.2 (Continuous: bulb lifetimes)**.** Type A (π_1=0.7): Exp(mean 2000h); Type B (π_2=0.3): Exp(mean 3000h). Given independent lifetimes X_1=2500, X_2=2700 (sum $T = 5200$), the joint densities are $f_k(x_1, x_2) = \lambda_k^2 e^{-\lambda_k(x_1+x_2)}$. The general decision boundary, setting $\pi_1 f_1(x) = \pi_2 f_2(x)$ and solving for $T = x_1 + x_2$, is linear in T:

$$T^* = 6000 \ln\left(\frac{\pi_1/2000^2}{\pi_2/3000^2}\right) = 6000 \ln(5.25) \approx 9949.3 \text{ hours},$$

classify as A if $T \le T^*$. Since $5200 \le 9949.3$, classify as Type A ($\pi_1 f_1/\pi_2 f_2 \approx 2.2 > 1$, consistent).

16.3 Bounds and attainability of the Bayes risk

Proposition 16.3. $R(\delta^*) \le \min(\pi_1, \pi_2) \le \frac{1}{2}$.

Proof

The constant rules $\delta_1 \equiv 1$, $\delta_2 \equiv 2$ have risks $R(\delta_1) = \pi_2$, $R(\delta_2) = \pi_1$. Since δ^* minimizes risk over *all* rules, $R(\delta^*) \le \min(R(\delta_1), R(\delta_2)) = \min(\pi_1, \pi_2) \le 1/2$ (the last inequality since $\pi_1+\pi_2 = 1$ is maximized at $\pi_1 = \pi_2 = 1/2$).

Equality is attainable when either (i) $f_1 = f_2$ a.e. (no discriminating information; δ^* defaults to the majority class, $R(\delta^*) = \min(\pi_1, \pi_2)$), or (ii) the density ratio is uniformly bounded, $\sup_x f_1(x)/f_2(x) \le k_0$, and $\pi_1 k_0 \le \pi_2$, forcing $\pi_1 f_1(x) \le \pi_2 f_2(x)$ everywhere so $\delta^* \equiv 2$, giving $R(\delta^*) = \pi_1 = \min(\pi_1, \pi_2)$.

16.4 Uniqueness of the Bayes rule

The Bayes rule is unique (up to null sets) iff $P(\{x : \pi_1 f_1(x) = \pi_2 f_2(x)\}) = 0$. If this "tie set" has positive probability, any allocation of it between the classes achieves the identical minimal risk.

Example 16.4 (Non-uniqueness)**.** Let $X \mid C=1 \sim \text{Unif}(0,1)$, $X \mid C=2 \sim \text{Unif}(0,2)$, $\pi_1 = 1/3, \pi_2 = 2/3$. On $(0,1)$: $\pi_1 f_1(x) = \frac{1}{3} = \pi_2 f_2(x)$ (an indifference region of positive measure); on $(1,2)$: $\pi_1 f_1 = 0 < \pi_2 f_2 = 1/3$, so every point there must go to class 2. For any threshold rule $\delta_c(x) = \mathbb{1}\{x < c\} \cdot 1 + \mathbb{1}\{x \ge c\} \cdot 2$ with $c \in (0,1]$, one computes $R(\delta_c) = \frac{1}{3}(1-c) + \frac{2}{3} \cdot \frac{c}{2} = \frac{1}{3}$ for *every* such c — yet $P(\delta_{c_1}(X) \ne \delta_{c_2}(X)) = \frac{2}{3}|c_2 - c_1| > 0$ for $c_1 \ne c_2$: distinct classifiers, identical minimal risk, hence non-uniqueness.

17 Gaussian Discriminant Analysis and the Mahalanobis Distance

17.1 Setup and the linear discriminant rule

Suppose $X \mid C=k \sim N_d(\mu_k, \Sigma)$ $(k = 1, 2)$ with a *common* covariance $\Sigma \succ 0$. The Bayes rule chooses class 1 iff $\ln(\pi_1 f_1(x)/\pi_2 f_2(x)) > 0$:

$$\ln \pi_1 - \tfrac{1}{2}(x - \mu_1)'\Sigma^{-1}(x - \mu_1) > \ln \pi_2 - \tfrac{1}{2}(x - \mu_2)'\Sigma^{-1}(x - \mu_2).$$

Expanding $(x - \mu_k)'\Sigma^{-1}(x - \mu_k) = x'\Sigma^{-1}x - 2\mu_k'\Sigma^{-1}x + \mu_k'\Sigma^{-1}\mu_k$, the quadratic term $x'\Sigma^{-1}x$ cancels (identical for both classes), leaving the **linear** decision rule

$$x'\Sigma^{-1}(\mu_1 - \mu_2) - \tfrac{1}{2}\mu_1'\Sigma^{-1}\mu_1 + \tfrac{1}{2}\mu_2'\Sigma^{-1}\mu_2 + \ln(\pi_1/\pi_2) \gtrless 0 \qquad \textbf{(LDA boundary)}.$$

This cancellation of the quadratic term is precisely *why* LDA is linear: with unequal covariances $\Sigma_1 \neq \Sigma_2$, the quadratic terms would not cancel, giving **Quadratic Discriminant Analysis (QDA)** instead.

17.2 Risk as a function of the Mahalanobis distance

Definition 17.1. The Mahalanobis distance between μ_1, μ_2 relative to shared Σ is $\Delta = \sqrt{(\mu_1 - \mu_2)'\Sigma^{-1}(\mu_1 - \mu_2)}$.

Under equal priors $\pi_1 = \pi_2 = \frac{1}{2}$, the Bayes risk is $R(\delta^*) = \Phi(-\Delta/2)$, where Φ is the standard normal CDF — strictly decreasing in Δ.

Proof

With $\pi_1 = \pi_2$, $\ln(\pi_1/\pi_2) = 0$ and the rule simplifies to $(\mu_1 - \mu_2)'\Sigma^{-1}\big(x - \frac{\mu_1+\mu_2}{2}\big) > 0$. Define $a = \Sigma^{-1}(\mu_1 - \mu_2)$ and $W = a'X$; the rule is $W > \gamma$, $\gamma = \frac{1}{2}a'(\mu_1 + \mu_2)$. Since X is Gaussian, W is univariate Gaussian with

$$\mathbb{E}[W \mid C{=}k] = a'\mu_k, \qquad \mathrm{Var}(W \mid C{=}k) = a'\Sigma a = (\mu_1 - \mu_2)'\Sigma^{-1}(\mu_1 - \mu_2) = \Delta^2 \quad (k = 1, 2).$$

Type I error (C=1 misclassified as 2): $P(W \le \gamma \mid C{=}1) = \Phi\big(\frac{\gamma - a'\mu_1}{\Delta}\big)$. Now $\gamma - a'\mu_1 = \frac{1}{2}a'(\mu_2 - \mu_1) = -\frac{1}{2}\Delta^2$, so this equals $\Phi(-\Delta/2)$. **Type II error** (C=2 misclassified as 1): $P(W > \gamma \mid C{=}2) = 1 - \Phi\big(\frac{\gamma - a'\mu_2}{\Delta}\big)$; here $\gamma - a'\mu_2 = \frac{1}{2}\Delta^2$, giving $1 - \Phi(\Delta/2) = \Phi(-\Delta/2)$ by symmetry of Φ. Weighting by equal priors, $R(\delta^*) = \frac{1}{2}\Phi(-\Delta/2) + \frac{1}{2}\Phi(-\Delta/2) = \Phi(-\Delta/2)$. Since $\Phi'(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2} > 0$, $\frac{d}{d\Delta}\Phi(-\Delta/2) = -\frac{1}{2}\Phi'(-\Delta/2) < 0$: strictly decreasing in Δ.

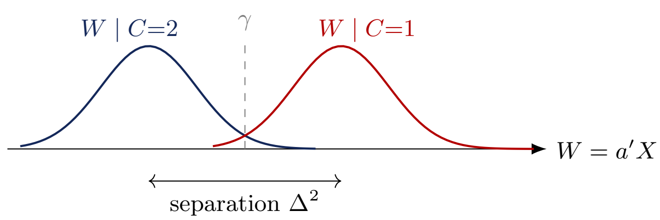

Figure 10: Projected class-conditional distributions of $W = a'X$; overlap area corresponds to $\Phi(-\Delta/2)$.

Setting	Decision criterion	Optimal boundary	Theoretical risk
General 0–1 loss	$\pi_1 f_1(x) \gtrless \pi_2 f_2(x)$	$\pi_1 f_1 = \pi_2 f_2$	$\pi_1 + \int_{R_1}[\pi_2 f_2 - \pi_1 f_1]\,dx \le \min(\pi_1, \pi_2)$
Identical densities	$\pi_1 \gtrless \pi_2$ everywhere	indifferent	$\min(\pi_1, \pi_2)$ (upper bound attained)
Overlapping uniforms	non-unique over tie region	indifference set, positive measure	unique min. risk, non-unique rule
Gaussian, equal Σ	$(\mu_1{-}\mu_2)'\Sigma^{-1}x \gtrless \gamma$	linear hyperplane in $\mathbb{R}^d$	$\Phi(-\Delta/2)$ for $\pi_1{=}\pi_2{=}\frac{1}{2}$, decreasing in Δ

17.3 Comparison with the parametric regression framework

For continuous $Y = \alpha + \beta X + \varepsilon$ under squared error loss, the theoretical risk $R(f) = \mathbb{E}[(Y - \alpha - \beta X)^2]$ is smooth, quadratic, convex in (α, β), with closed-form OLS minimizer. This contrasts sharply with 0–1 classification loss, whose empirical version is discontinuous and non-convex — a structural reason why (a) Gaussian-model-based classifiers (LDA/QDA) and (b) convex surrogate-loss classifiers (logistic regression, SVM) are the two dominant practical strategies.

17.4 Linear classifiers cannot solve the checkerboard (parity) problem

Let $(X_1, X_2) \in \{1, \ldots, K\}^2$ on a grid with $C = 1$ if $X_1 + X_2$ even, $C = 2$ if odd. Any linear boundary $\alpha + \beta_1 X_1 + \beta_2 X_2 = 0$ achieves $\approx 50\%$ error (no better than guessing), and simple linear regression fails identically because the marginal correlation of either coordinate with C is exactly zero by the grid's symmetry. Resolving parity requires **local recursive partitioning** (deep trees) or **higher-order tensor-product interactions** (MARS with interaction terms, or projections along non-axis-aligned directions as in PPR) — a concrete illustration of why purely additive / purely linear methods are fundamentally limited, motivating the tree- and projection-based methods of Parts II–III.

Part V: Robust Regression — OLS vs. LAD and IRWLS

18 Two Loss Functions as M-Estimators

Consider (initially, through the origin) $Y_i = \beta X_i + \varepsilon_i$. Both OLS and LAD are **M-estimators**: minimize $\sum_i \rho(r_i)$, $r_i = Y_i - \beta X_i$, for a choice of ρ.

	OLS	LAD
$\rho(r)$	r^2	$\lvert r\rvert$
$\psi(r) = \rho'(r)$	$2r$ (unbounded)	$\mathrm{sign}(r)$ (bounded, undefined/0 at $r = 0$)

The organizing idea

ψ is the *influence* of a single residual on the fit. OLS's influence grows without bound as $|r| \to \infty$; LAD's saturates at ± 1. Robustness (to Y-space contamination) is entirely a statement about the boundedness of ψ.

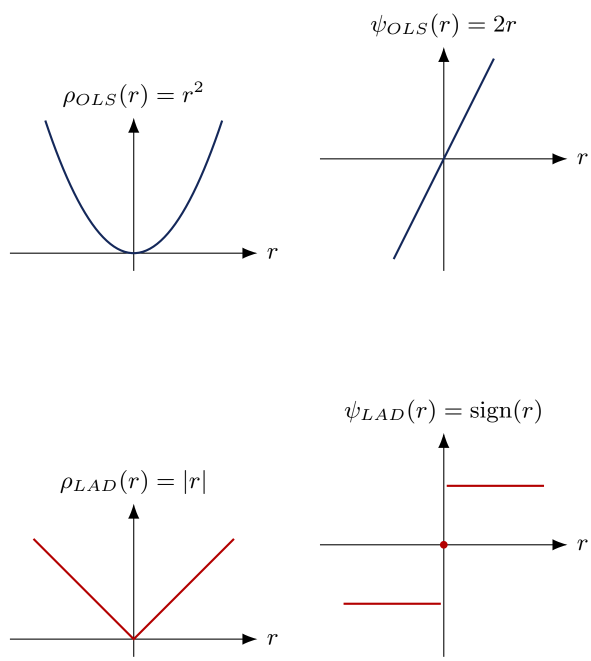

Figure 11: ρ and $\psi = \rho'$ for OLS (top) and LAD (bottom): OLS's influence is unbounded, LAD's saturates.

19 Estimators Without an Intercept

OLS has the familiar closed form $\widehat{\beta}_{OLS} = \sum_i X_i Y_i / \sum_i X_i^2$. **LAD** minimizes $\sum_i |Y_i - \beta X_i| = \sum_i |X_i|\,|r_i - \beta|$ where $r_i = Y_i/X_i$ ($X_i \neq 0$) — exactly the objective minimized by a **weighted median** of $\{r_i\}$ with weights $\{|X_i|\}$:

$$\widehat{\beta}_{LAD} = \text{weighted median of } \{Y_i/X_i\} \text{ with weights } \{|X_i|\}.$$

Proof

Why the weighted median solves $\min_\beta \sum_i w_i |r_i - \beta|$**.** The objective is convex and piecewise linear in β; its subgradient is $g(\beta) = -\sum_{i:r_i>\beta} w_i + \sum_{i:r_i<\beta} w_i + [-w_{i_0}, w_{i_0}]$ if $\beta = r_{i_0}$ for some tied index. As β increases past each r_i, that term's contribution flips sign, decreasing the "upper weight" and increasing the "lower weight" by w_i. The minimizer is any β where cumulative weight below β first reaches (at least) half the total weight $\sum_i w_i$ — exactly the definition of the weighted median. Formally, 0 lies in the subgradient set at β^* iff $\sum_{i:r_i<\beta^*} w_i \le \frac{1}{2}\sum_i w_i$ and $\sum_{i:r_i>\beta^*} w_i \le \frac{1}{2}\sum_i w_i$, which is precisely the weighted-median condition.

No closed form exists, but it is exact and cheap: sort $\{r_i\}$, accumulate weight, stop at the first point cumulative weight reaches 50% of the total.

19.1 Skewed errors: mean vs. median (not a robustness phenomenon)

Even with *no* outliers, OLS and LAD can disagree if ε is skewed but still mean-zero, e.g. $\varepsilon \sim \chi^2_k - k$. OLS targets $\mathbb{E}[Y \mid X] = \beta X$ (exact, since $\mathbb{E}[\varepsilon] = 0$); LAD targets $\text{median}(Y \mid X) = \beta X + \text{median}(\varepsilon)$. Since χ^2_k is right-skewed, $\text{median}(\chi^2_k) < k = \mathbb{E}[\chi^2_k]$, so $\text{median}(\varepsilon) < 0$ and LAD's fitted line sits systematically *below* OLS's — a consequence of $\mathbb{E}[\varepsilon] \ne \text{median}(\varepsilon)$ under skewness, conceptually distinct from robustness to contamination.

19.2 Y-space vs. X-space outliers

Y-space outlier (ordinary X, wildly wrong Y): OLS's squared residual from this point grows quadratically, dragging the fit toward it; LAD's influence saturates at ± 1 regardless of how extreme the residual is, so the LAD fit barely moves.
Proposition 19.1 (Influence function of $\widehat{\beta}_{OLS}$)**.** *Perturbing $Y_j \to Y_j + \delta$ in the no-intercept model, $\partial\widehat{\beta}_{OLS}/\partial\delta = X_j/\sum_i X_i^2$ is a fixed nonzero* constant *in δ (linear influence), so the total shift $\delta \cdot X_j/\sum_i X_i^2$ grows without bound as $\delta \to \infty$: OLS has unbounded influence. LAD's corresponding shift is bounded because its ψ saturates.*

Proof

$\widehat{\beta}_{OLS} = \sum_i X_i Y_i / \sum_i X_i^2$; perturbing only Y_j, $\partial\widehat{\beta}_{OLS}/\partial\delta = X_j/\sum_i X_i^2$, a fixed nonzero constant, so the shift in $\widehat{\beta}_{OLS}$ is exactly *linear* in δ and grows without bound as $\delta \to \infty$ — unlike LAD, whose weighted-median solution moves by at most one "rank position" regardless of how large δ is, since ψ_{LAD} has already saturated at ± 1.

X-space outlier (high leverage, i.e. extreme X, with an inconsistent Y): *both* estimators break. LAD's weights are $w_i = |X_i|$: an extreme X_i dominates the total weight $\sum_i |X_i|$, so as $X_i \to \infty$ for one point, the weighted median is forced toward Y_i/X_i for that point alone — *independent of how wrong Y_i is.* **Key message: robustness in Y-space and X-space are separate properties; LAD only buys the former.**

20 Adding an Intercept

For $Y_i = \alpha + \beta X_i + \varepsilon_i$: OLS retains its closed form $\hat{\theta}_{OLS} = (X^\top X)^{-1} X^\top y$, $\theta = (\alpha, \beta)^\top$. LAD's first-order (subgradient) conditions for $\min_{\alpha,\beta} \sum_i |Y_i - \alpha - \beta X_i|$ are the coupled, non-smooth system

$$\sum_{i=1}^n \text{sign}(Y_i - \alpha - \beta X_i) = 0, \qquad \sum_{i=1}^n X_i \text{sign}(Y_i - \alpha - \beta X_i) = 0.$$

The one-dimensional weighted-median trick does not directly extend to two simultaneous unknowns — the entry point for **IRWLS**.

21 Iteratively Reweighted Least Squares (IRWLS)

21.1 The algebraic trick

For any $r_i \ne 0$, $|r_i| = r_i^2/|r_i|$. If we *freeze* $w_i = 1/|r_i|$ at the current fit's residuals, then $\sum_i |r_i| = \sum_i w_i r_i^2$ exactly (not an approximation, whenever $r_i \ne 0$). Freezing w_i turns the non-smooth LAD objective into an ordinary (smooth, quadratic) **weighted least squares** problem at each iteration; since the weights depend on the residuals, which depend on (α, β), we alternate.

21.2 The algorithm

1. **Initialize** $\theta^{(0)} = \hat{\theta}_{OLS} = (X^\top X)^{-1} X^\top y$.
2. **Residuals**: $r^{(t)} = y - X\theta^{(t)}$.
3. **Reweight**: $w_i^{(t)} = 1/\max(|r_i^{(t)}|, \delta)$ for a small floor $\delta > 0$ (guards $r_i^{(t)} \approx 0$); $W^{(t)} = \mathrm{diag}(w_1^{(t)}, \ldots, w_n^{(t)})$.
4. **Weighted least squares**: $\theta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} y$.
5. **Repeat** until $\|\theta^{(t+1)} - \theta^{(t)}\| < \varepsilon$.

Reading the weight

A point with a small residual (the fit already passes near it) gets a *large* weight $w_i = 1/|r_i|$; a point with a large residual (possible outlier) gets a *small* weight. This is why IRWLS converges toward the LAD solution: it iteratively down-weights points the current fit disagrees with most, exactly mirroring LAD's saturating influence function. If $r_i^{(t)} = 0$ exactly, the floor δ prevents $w_i \to \infty$; without it the weight would blow up, causing numerical instability (the point would completely dominate the next WLS step).

Proposition 21.1. *For a general M-estimator with loss ρ, the IRWLS weight is $w_i = \rho'(|r_i|)/|r_i|$.*

Proof

Setting $\partial_\beta \sum_i \rho(r_i) = 0$ gives $\sum_i \rho'(r_i)\, \partial r_i/\partial\beta = 0$; writing $\rho'(r_i) = \left[\rho'(r_i)/r_i\right] \cdot r_i$ and identifying the bracket as w_i recasts the stationarity condition as the normal equations of a weighted least-squares problem $\sum_i w_i r_i\, \partial r_i/\partial\beta = 0$. For $\rho_{OLS}(r) = r^2$: $\rho'(r) = 2r \Rightarrow w_i = 2r_i/r_i = 2 \equiv$ const — unweighted OLS is the (trivial) fixed point of its own IRWLS iteration. For $\rho_{LAD}(r) = |r|$: $\rho'(r) = \mathrm{sign}(r) \Rightarrow w_i = \mathrm{sign}(r_i)/r_i = 1/|r_i|$, matching Step 3 above. This generalizes immediately to Huber's loss (quadratic near 0, linear in the tails) and Tukey's bisquare (redescending, weight $\to 0$ for very large $|r_i|$), which further bound the influence of extreme points beyond what LAD achieves.

21.3 Convergence trace and order

A representative $n = 12$ run with true $(\alpha, \beta) = (3, 2.5)$: The ratio $\|\Delta\theta^{(t+1)}\|/\|\Delta\theta^{(t)}\|$ decays roughly geometrically once

t	$\alpha^{(t)}$	$\beta^{(t)}$	$\|\theta^{(t)} - \theta^{(t-1)}\|$
0 (OLS init)	3.6429	2.4660	–
1	3.2393	2.5260	4.08×10^{-1}
2	3.2069	2.5329	3.31×10^{-2}
5	3.0855	2.5486	4.50×10^{-2}
10	2.9117	2.5684	1.96×10^{-2}
15	2.8904	2.5709	4.60×10^{-4}
20	2.8902	2.5709	4.88×10^{-6}
24 (converged)	2.8901	2.5709	4.18×10^{-13}

near the fixed point (e.g. from t=15 to t=24 the norm shrinks by orders of magnitude each iteration) — behaviour consistent with (at best) **linear** convergence of IRWLS as a fixed-point/majorize-minimize iteration (each step exactly minimizes a quadratic *majorizer* of the LAD objective, a general property of MM algorithms, which typically converge linearly rather than at the quadratic rate of, say, Newton's method).

22 An Alternative: Alternating Conditional Medians

Both one-dimensional conditional sub-problems of LAD-with-intercept have closed forms: fixing β, $\min_\alpha \sum_i |(Y_i - \beta X_i) - \alpha| \Rightarrow \hat{\alpha} = \text{median}(Y_i - \beta X_i)$; fixing α, $\min_\beta \sum_i |(Y_i - \alpha) - \beta X_i| \Rightarrow \hat{\beta} =$ weighted median of $\{(Y_i - \alpha)/X_i\}$ with weights $\{|X_i|\}$ (Section 2's trick, applied to $Y_i - \alpha$). One can alternate these two exact 1-D minimizations (coordinate descent). Because $\sum_i |Y_i - \alpha - \beta X_i|$ is **jointly convex** in (α, β) (a sum of convex functions of an affine map of (α, β)), and each coordinate update is an *exact* minimization along its coordinate, this coordinate descent is guaranteed to converge to the joint minimizer for smooth strictly convex objectives; for non-smooth but jointly convex objectives, coordinate descent can in general stall at a non-optimal point where no single coordinate move improves the objective but a joint move would (this can happen when the non-smooth "kinks" of different coordinates interact). In practice, for the LAD objective, this failure mode is rare but not theoretically excluded — which is one reason IRWLS (a joint, not coordinate-wise, majorize-minimize step) is generally preferred.

23 Open Questions and Extensions

- **Leverage-blind weighting**: IRWLS's weight $w_i = 1/|r_i|$ downweights large *residuals*, but a high-leverage point can pull the whole line toward itself, leaving its *own* residual small — so IRWLS (like plain LAD) does not fully solve the X-space outlier problem. A leverage-aware fix downweights by a function of the hat-matrix diagonal h_{ii} as well (Mallows-/Huber-type bounded-influence M-estimators), the natural bridge to fully robust regression with bounded influence in *both* X and Y.
- **Cauchy errors**: Cauchy errors have no finite mean or variance but a well-defined median = 0; OLS is ill-behaved (infinite-variance target), while LAD (targeting the median) remains well-behaved, and IRWLS still converges to it.
- **Quantile regression**: LAD is the $\tau = 1/2$ case of the more general family $\rho_\tau(r) = r(\tau - \mathbb{1}\{r < 0\})$, targeting the τ-th conditional quantile of $Y \mid X$ instead of the median. At $\tau = 1/2$, $\rho_{1/2}(r) = \frac{1}{2}|r|$, recovering ρ_{LAD} up to the constant $\frac{1}{2}$; the corresponding IRWLS weight for general τ is $w_i = \rho'_\tau(|r_i|)/|r_i|$, giving an asymmetric weighting scheme that under- or over-weights positive vs. negative residuals depending on τ. Same M-estimator machinery, same IRWLS-style fitting, different ρ.

Appendix: Consolidated Summary

A Grand Comparison of Methods

Method	Form of $\hat{f}(x)$ / rule	Flexibility source / key idea
Linear model (OLS)	$w_0 + \sum_i w_i x_i$	baseline global linear fit
Ridge	$(X'X + \lambda I)^{-1} X'y$	isotropic ℓ_2 shrinkage, no selection
LASSO	soft-threshold of OLS (orthonormal case)	ℓ_1 shrinkage *and* exact sparsity
Elastic Net	threshold + shrink, correlated grouping	$\ell_1 + \ell_2$ mix; sparse *and* stable under collinearity
Regression splines	$\sum_j \alpha_j B_j(x)$, truncated-power/B-spline basis	local polynomial pieces, continuity at knots
Additive model	$\sum_j f_j(x_j)$	sum of univariate (spline) effects, no interactions
Projection pursuit	$\sum_k f_k(\alpha'_k x)$	ridge functions along estimated directions
Single-layer perceptron / GLM	$\phi(w_0 + \sum_i w_i x_i)$, ϕ fixed	known link on a linear index
MLP (1 hidden layer)	$f\{w_0 + \sum_j w_j f_j(w_{0j} + \sum_i w_{ij} x_i)\}$	composed linear indices & fixed nonlinearities; universal approximator as $k \to \infty$
k-NN / kernel regression	local average of nearby Y_i's	purely local averaging, no global functional form
Regression tree	$\sum_m c_m \mathbb{1}\{x \in R_m\}$, adaptive rectangles	recursive binary splitting minimizing within-node variance
Bagging	$\frac{1}{B}\sum_b \hat{f}_b(x)$, bootstrap trees	variance reduction via averaging correlated estimators
Random forest	bagging + feature subsampling at splits	decorrelates trees, lowers the variance floor $\rho\sigma^2$
MARS	$\sum$ tensor-product hinge functions	continuous adaptive splines with interactions
LDA / QDA	posterior-probability threshold, Gaussian model	linear (equal Σ) or quadratic (unequal Σ) boundary
Bayes classifier	$\arg\max_k \pi_k f_k(x)$	risk-optimal rule given true densities
LAD / IRWLS	weighted median / iteratively reweighted LS	bounded influence, robust to Y-space outliers

Table 2: A birds-eye comparison of every method developed in these notes.

B Notation

- $\boldsymbol{X} \in \mathbb{R}^{n\times(p+1)}$: design matrix (first column of ones for the intercept); $\boldsymbol{X}_c$: column-centered design (no intercept column).
- $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$ (or $\mathbb{R}^p$ centered): coefficient vector; $\widehat{\beta}$: an estimator of $\boldsymbol{\beta}$.
- λ, α, h: tuning/regularization parameters (ridge/LASSO penalty, elastic-net mixing weight, kernel bandwidth) — always chosen by cross-validation unless stated otherwise.
- $(u)_+ = \max(u, 0)$: positive-part / hinge function.
- $i(t)$: node impurity (variance for regression trees; Gini/entropy/misclassification for classification trees).
- π_k, f_k: class prior and class-conditional density for class k.
- $\Delta = \sqrt{(\mu_1 - \mu_2)'\Sigma^{-1}(\mu_1 - \mu_2)}$: Mahalanobis distance.

- $\rho, \psi = \rho'$: M-estimator loss and its derivative ("influence") function.

C Cross-Cutting Themes

Several unifying threads run through these notes and are worth stating explicitly:

1. **Bias–variance trade-off** appears in every method: the ridge/LASSO penalty λ, the number of spline knots or MARS basis functions, the network width k, the neighborhood size k in k-NN or bandwidth h in kernel regression, and the tree-pruning parameter α are all, at heart, the *same* knob — trading fit (low bias) against stability (low variance) — and are all tuned the same way, by cross-validation.
2. **Nested model sequences and greedy search** recur in subset selection (§1.2), knot selection for splines (§6.5), forward/backward MARS (§17), and tree pruning (§9): build a sequence of candidate models differing by one structural unit, then select via a criterion (adjusted R^2, AIC/BIC, cross-validated error, or GCV).
3. **From additive to interactive**: additive splines cannot capture x_1x_2-type interactions (§6.6); projection pursuit and MARS both resolve this, respectively by rotating the projection direction or by taking tensor products of hinge functions — the same limitation (and the same two classes of fix) that makes purely linear classifiers fail on the checkerboard problem (§22.6).
4. **Ensembling reduces variance but is capped by correlation**: bagging's variance is $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ (§13.4); more trees alone cannot beat the correlation floor $\rho\sigma^2$, motivating random forests' explicit decorrelation via feature subsampling.
5. **Robustness is about bounded influence**: the entire distinction between OLS and LAD, and the entire mechanism of IRWLS and its generalizations (Huber, Tukey bisquare, quantile regression), reduces to the shape of $\psi = \rho'$ — bounded vs. unbounded — exactly mirroring how LASSO's non-smooth penalty (vs. ridge's smooth one) is what enables exact variable selection.